

Warehouses and Diplomas:

Amazon's Expansion and Local Educational Attainment

Md Shafiqul Islam*

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Abstract

This paper presents a comprehensive analysis of how Amazon's logistics facility rollout affected human capital formation. I exploit variation in the timing of Amazon's entry across commuting zones between 2010 and 2024. The rollout increased educational attainment, especially bachelor's and master's degrees. More graduates moved to or stayed in these areas. Part-time enrollment declined while full-time enrollment increased. These outcomes likely reflect spillover effects from Amazon's entry. Amazon's presence expanded skilled employment at other firms and increased demand for college graduates.

Keywords: Amazon; e-commerce; educational attainment; college enrollment; human capital; local labor markets; difference-in-differences

JEL codes: I24; I25; J24; L81; R23

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1 Introduction

Over the past fifteen years, the United States has seen substantial growth in low-skill employment, with Amazon emerging as a major contributor.¹ Amazon expanded into local labor markets nationwide, supported by billions of dollars in state and local subsidies ([Good Jobs First, 2022](#)). The company's warehouses employ a large proportion of individuals without college degrees ([Cunningham, 2024](#)).

The entry of a new employer into a local labor market that offers jobs without a degree often reshapes both educational and employment dynamics. Young individuals face a clear trade-off between pursuing further education and accepting immediate employment opportunities. The existing literature on economic booms indicates that many young people choose to enter the workforce. For example, coal, fracking, and housing booms have raised wages for low-skill labor and led teenagers and young adults to leave school ([Black et al., 2005](#); [Cascio and Narayan, 2022](#); [Charles et al., 2018](#)). In Mexico, the expansion of export manufacturing similarly increased school dropout rates among youth who were approximately 16 years old when new factory jobs became available ([Atkin, 2016](#)).

The job growth in Amazon's warehouses follows the same historical patterns seen in previous job booms. In addition to operating warehouses, Amazon provides logistics services to other firms, and its facilities attract suppliers, network sellers, and business services that employ engineers, managers, and other college graduates. Large plant openings create positive spillovers on other firms ([Greenstone et al., 2010](#)), and new jobs in traded industries generate additional local jobs, especially skilled ones ([Moretti, 2010](#)). The net effect of Amazon's entry on human capital remains theoretically ambiguous. Warehouse employment raises the opportunity cost of staying in school, whereas skilled positions created through spillovers at other firms, as well as a limited number at Amazon, increase the returns to educational attainment and enrollment. Existing studies find that Amazon's entry raises local employment and income ([Pathania and Netessine, 2026](#); [Cunningham, 2024](#)), but they do not examine whether it changes the education of the local population. This paper exploits the staggered timing of Amazon's entry into U.S. commuting zones between 2010 and 2024 to show that the rise in the expected return to a degree outweighs the rise in the opportunity cost of schooling, as Amazon's arrival substantially raises the share of adults with a bachelor's or master's degree.

I conduct a comprehensive analysis of the impact of Amazon's expansion on human capital

¹Employment in warehousing and storage grew from about 410,000 workers in 1990 to 630,000 in 2010 and then nearly tripled, peaking at 1.9 million in 2022 (Bureau of Labor Statistics Current Employment Statistics, NAICS 493, series CES4349300001, seasonally adjusted). Over the same years, Amazon went from about 20 U.S. facilities to nearly 1,400 (MWPVL International; fulfillment centers, sortation centers, delivery stations, and air hubs). Appendix Figure A1 shows that the two took off together after 2010.

formation in the United States. I match the opening dates and locations of all Amazon facilities, from MWPVL International ([MWPVL International, 2025](#)), with a panel of 297 U.S. commuting zones, 188 of which Amazon entered between 2010 and 2024. Educational attainment comes from published American Community Survey 1-year estimates for counties with at least 65,000 residents, 2005–2024, and IPUMS microdata add income, occupations, and migration. I measure college enrollment with the Integrated Postsecondary Education Data System and labor demand with County Business Patterns, the Business Dynamics Statistics, the Quarterly Workforce Indicators, and online job postings.²

Amazon strategically selects locations that align with its logistics network requirements, including proximity to major consumer markets and highways ([Houde et al., 2023](#)). The company targeted larger, more prosperous, and more educated commuting zones, so a comparison of places with and without Amazon would be misleading. I compare commuting zones that Amazon has already entered with those it enters later, employing the staggered difference-in-differences estimator of [Callaway and Sant’Anna \(2021\)](#). The analysis relies on the parallel trends assumption, which posits that, absent Amazon’s entry, educational attainment in early and later entrant zones would have followed similar trajectories. Prior to Amazon’s arrival, the shares of adults holding a bachelor’s or master’s degree increased at comparable rates in both groups, and the event-study coefficients before entry are close to zero.

I find that Amazon’s entry substantially raises local human capital. The share of adults aged 25 and older with a bachelor’s degree rises by 0.54 percentage points, about 2 percent of its pre-entry level, and the effect keeps growing, reaching 1.3 percentage points six years after entry. The share with a master’s degree rises by 0.42 percentage points and by about 0.8 percentage points after six years. The shares with a high school diploma or some college do not change. These gains could come from residents earning more degrees or from degree holders moving in, and the evidence points to both. Amazon’s entry attracts educated movers, since among people who moved from another state or from abroad in the past year, the number with a bachelor’s degree rises by about 7 percent and the number with a master’s degree by about 11 percent, while movers without a degree do not change. Even if all new graduates who moved in stayed, they would account for at most two-thirds of the increase in the bachelor’s share, so at least a third comes from people already living in these areas, including residents who earn degrees.

I measure first-time enrollment following [Charles et al. \(2018\)](#) and exclude colleges that drew 30 percent or more of their first-time students from other states or abroad. After Amazon’s arrival, the share of local college students enrolled part time falls by 1.6 percentage points, and first-time

²The Quarterly Workforce Indicators ([U.S. Census Bureau, 2026](#)) report industry employment from state unemployment-insurance records, 2005–2023. The IRS Statistics of Income migration data provide county-to-county migration flows and the income of movers, 2011–2023. Online job postings are from Lightcast (formerly Emsi Burning Glass), 2014–2023, using the county-level data of [Andreadis et al. \(2025\)](#).

enrollment at local four-year colleges rises by 0.26 percentage points of the population aged 18–24, about 5 percent of its pre-entry level. This estimate is less precise than the attainment results. Because the enrollment data do not track individual students, I cannot tell whether the same students shift from part-time to full-time study, and I view these findings as suggestive of a shift toward full-time study at four-year colleges.

I present additional evidence to support my claim that Amazon’s entry increases human capital formation. I validate the main result with randomization inference, placebo entry dates, a sample that ends before the pandemic, a balanced event window, and an alternative estimator. Randomly reassigning Amazon’s entry years across commuting zones almost never produces an effect as large as the observed one.³ Moving the entry forward by five years yields only minor placebo effects, and they are not statistically significant. Restricting the sample to data ending in 2019, prior to the pandemic, gives similar results.⁴ Keeping only commuting zones observed for all seven years after entry gives effects of 0.62 percentage points on the bachelor’s share and 0.42 percentage points on the master’s share. The synthetic difference-in-differences estimator of [Arkhangelsky et al. \(2021\)](#) estimates effects of 0.67 percentage points on the bachelor’s share and 0.45 percentage points on the master’s share.

Why does Amazon’s arrival raise education when earlier booms lowered it? I examine two channels, income and the demand for skilled labor. Median household income rises by 1.5 percent, or about \$960 per household in 2023 dollars, but rents rise faster. The income and education gains are concentrated in commuting zones that were already better off, not in poorer places where financial constraints bind most, so higher income is unlikely to explain the rise in attainment. The evidence points instead to higher demand for skilled labor. After Amazon arrives, professional, scientific, and technical establishments grow by 3.9 percent, employment in the information sector grows by about 9 percent, and the share of job postings for artificial intelligence and other advanced-technology jobs rises. Local employment shifts toward STEM occupations, especially computer and mathematical jobs, whose share rises by about 12 percent, and the share of jobs held by college graduates rises by 0.7 percentage points. Much of this demand comes from employers other than Amazon. The college wage premium does not change. Amazon’s spillovers add skilled jobs close to home without raising their pay, and residents appear to respond by investing in a college education while graduates stay in the area.

My work is related to three distinct literatures. First, existing studies on local labor demand shocks demonstrate that increases in low-skill employment frequently lead to reduced educational attainment ([Black et al., 2005](#); [Cascio and Narayan, 2022](#); [James, 2017](#); [Charles et al., 2018](#); [Atkin,](#)

³At most one of 500 random assignments produces an effect as large as the actual one (randomization-inference $p \leq 0.002$).

⁴The randomization-inference and pre-pandemic checks use a second panel built from IPUMS ACS microdata, which covers all commuting zones, including those without a county of at least 65,000 residents.

2016). In contrast, when economic expansion is driven by a single large employer, educational attainment may rise if the resulting growth generates additional skilled jobs at other firms. This complements evidence from India, where the arrival of information-technology service jobs raised local school enrollment (Shastri, 2012; Oster and Steinberg, 2013). Therefore, the specific types of jobs created by such shocks are crucial for determining educational outcomes.⁵ Second, these insights inform the evaluation of place-based policies and the incentives that local governments offer to large employers (Busso et al., 2013; Kline and Moretti, 2014; Greenstone et al., 2010; Bartik, 2020), which often neglect human capital considerations. The rationale for such incentives depends in part on their long-term effects on local workers (Bartik, 2020; Rossi-Hansberg et al., 2021), and these effects may be shaped as much by the types of jobs created indirectly as by those offered directly. Finally, this paper builds on recent research regarding Amazon’s local impact (Pathania and Netessine, 2026; Cunningham, 2024) and on worker responses to local demand shocks through migration (Notowidigdo, 2020; Monras, 2018; Wilson, 2022), by showing that Amazon’s entry affects both degree attainment and the residential decisions of degree holders.

Section 2 describes Amazon’s distribution network and presents a simple conceptual framework of schooling choice in a local labor market. Section 3 introduces the data and the empirical strategy. Section 4 investigates the impact of Amazon’s entry on educational attainment, migration, and college enrollment. Section 5 explores why Amazon’s entry raises education, examining household income and the demand for skilled work. Section 6 validates the results through placebo tests, randomization inference, and alternative estimators. Finally, Section 7 discusses policy implications and concludes.

2 Background and Conceptual Framework

2.1 Amazon’s Distribution Network

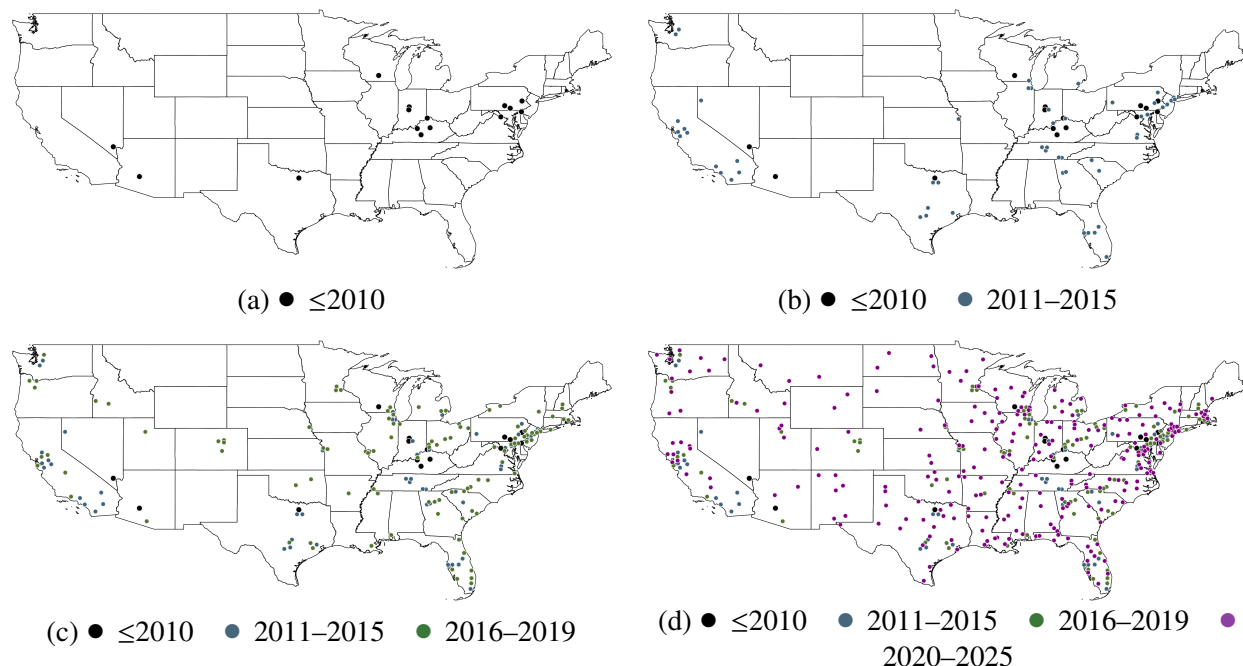
Amazon’s logistics network has three layers. Fulfillment centers store inventory and pick and pack orders, sortation centers group packages by destination, and delivery stations handle the last mile to customers (Rodrigue, 2020; Cunningham, 2024). Air hubs and delivery stations for heavy and bulky items complete the network. Amazon operated a small number of fulfillment centers in 2010 and more than one thousand facilities by the early 2020s (Figure 1). The period I study, 2010 through 2024, covers the fastest phase of this expansion.

Amazon’s location decisions changed during this period. Through the 2000s, the company placed fulfillment centers in a small number of low-population states, in part to avoid collecting

⁵Shocks that reduce demand for low-skill work can have similar effects. For instance, Chinese import competition increased high school graduation rates in the most affected U.S. labor markets (Greenland and Lopresti, 2016), and students’ field choices respond to local, rather than national, demand for their skills (Weinstein, 2022).

sales tax in the large states where most of its customers lived (Houde et al., 2023). Legal challenges and new state legislation made this strategy unsustainable. From about 2012, Amazon located facilities to minimize shipping costs and delivery times, which required building near customers in large, high-tax states (Houde et al., 2023; MWPVL International, 2025). The result was a staggered entry into local labor markets over more than a decade, which provides the variation in timing that this paper uses.

Figure 1: Staggered Rollout of Amazon’s Logistics Network by Year, 2005–2025



Notes: Each panel plots the cumulative distribution of the county locations of Amazon facilities of all types (fulfillment centers, sortation centers, delivery stations, and air hubs) by opening-year cohort: black dots, facilities opened by 2010; blue dots, 2011–2015; green dots, 2016–2019; purple dots, 2020–2025. Data compiled from MWPVL International facility-tracking records. Facility locations are shown at the county level; the empirical analysis is conducted at the commuting-zone level.

The treatment in this paper is Amazon’s entry into a commuting zone, defined as the opening of its first facility of any type. Entry usually brings more than one facility over time: of the 188 commuting zones Amazon entered, 59 received a fulfillment center in the year of entry and 112 had one by 2024, while in most of the rest the first facility was a delivery station or a sortation center. The estimates therefore capture the effect of Amazon’s arrival as a whole rather than of a single type of facility.⁶

⁶Amazon does not choose locations at random. Chung (2023) finds that fulfillment centers are more likely to open in counties with higher-than-average population density and education and lower-than-average median household income and median age, consistent with Amazon seeking large labor pools close to major consumer markets. Section 3.2 discusses how the design addresses this selection.

2.2 Conceptual Framework

This analysis is structured around a simple model of schooling choice in a local labor market, drawing on [Becker \(1964\)](#) and the college-enrollment framework of [Charles et al. \(2018\)](#). In this model, a young person i with academic ability $\theta_i \in [0, 1]$ selects among three options: enrolling in a four-year college ($c = B$), enrolling in a two-year college ($c = A$), or entering the workforce without pursuing further education. The net payoff from college type c is

$$R^c(\theta_i) = \mathbb{E}[\Pi_c] - F_c - \kappa_c(1 - \theta_i) - Y_{NC}, \quad (1)$$

where $\mathbb{E}[\Pi_c]$ is the expected lifetime earnings gain from degree c , F_c is the direct cost of attending, $\kappa_c(1 - \theta_i)$ is an effort cost that declines with ability, and Y_{NC} is the earnings forgone while in school, measured by the wages available to workers without a degree. A labor demand shock S , here the arrival of Amazon, changes the payoff by

$$\frac{dR^c(\theta_i)}{dS} = \underbrace{\frac{d\mathbb{E}[\Pi_c]}{dS}}_{\text{upgrading}} - \underbrace{\frac{dF_c}{dS}}_{\text{income}} - \underbrace{\frac{dY_{NC}}{dS}}_{\text{substitution}}. \quad (2)$$

Substitution reduces the economic return to obtaining a college degree. For example, Amazon warehouses employ individuals without a degree. When such employment opportunities increase wages or job availability, they raise Y_{NC} and consequently lower the relative payoff to college, a dynamic previously documented in the contexts of coal, fracking, housing, and export-manufacturing booms ([Black et al., 2005](#); [Cascio and Narayan, 2022](#); [Charles et al., 2018](#); [Atkin, 2016](#)). This substitution effect is most pronounced among students who are on the margin between entering the workforce and pursuing further education, particularly those who combine employment with part-time study at two-year colleges.

Upgrading operates in the opposite direction. The entry of Amazon generates positive spillover effects for other firms by attracting suppliers, logistics users, and business services that employ engineers, managers, and other college graduates ([Greenstone et al., 2010](#); [Rossi-Hansberg et al., 2021](#); [Moretti, 2010](#)). The expected gain from a degree can be expressed as the product of two terms:

$$\mathbb{E}[\Pi_B] = p_B \cdot \Delta w_B, \quad (3)$$

Here, p_B denotes the probability that a graduate secures employment in the local labor market utilizing the degree, and Δw_B represents the corresponding wage gain. A demand shock can increase the expected gain through either component. An increase in the college wage premium Δw_B raises the local return to a degree. Alternatively, if the number of skilled jobs increases, and thus p_B rises,

the return increases even if the wage premium remains unchanged. The latter scenario is particularly likely when college graduates are mobile, as new skilled jobs are often filled by graduates who remain in or relocate to the area, thereby preventing a rise in the local wage premium.

Household income influences the cost of college attendance. If Amazon’s entry increases household income and families experience borrowing constraints, the effective cost F_c decreases (Lochner and Monge-Naranjo, 2011; Lovenheim, 2011; Carneiro and Heckman, 2002; Johnson, 2013). This mechanism is likely to be most significant in areas where borrowing constraints are most severe, particularly in lower-income regions.

In addition to schooling decisions, the number of degree holders in a commuting zone may also change due to graduate migration. The same increase in skilled jobs that raises p_B for local students makes the area more attractive to graduates from other regions and reduces the incentive for current residents to leave (Notowidigdo, 2020; Monras, 2018). These two margins differ in timing: changes in schooling decisions affect the stock of adults with a degree only after the necessary years to complete the degree, whereas changes in location decisions can alter the stock almost immediately.

The framework yields four predictions that guide the empirical analysis. First, if upgrading exceeds substitution, both four-year enrollment and bachelor’s degree attainment are expected to rise, while enrollment among students who combine work with part-time or two-year study may decrease. Second, if upgrading is primarily driven by the availability of skilled jobs, the shares of employment in skilled industries and occupations should increase, although the college wage premium may remain stable. Third, if the income channel is significant, effects are more pronounced in areas where households had lower incomes prior to Amazon’s arrival. Fourth, if migration plays a significant role, the stock of degree holders will adjust more rapidly than can be explained by new enrollment alone. The net effect on educational attainment remains an empirical question.

3 Data and Empirical Strategy

3.1 Data

I combine data on the timing and location of Amazon’s facilities with data on the education, enrollment, and labor markets of the places they enter. Information on Amazon’s facilities comes from MWPVL International (MWPVL International, 2025), a supply-chain consulting firm that tracks Amazon’s logistics network. For each facility, the data record the address, opening date, facility type (fulfillment center, sortation center, delivery station, heavy and bulky delivery station, or air hub), and floor space. I use the five facility types in Amazon’s logistics network and exclude Amazon Fresh and Whole Foods grocery stores, which are retail outlets rather than logistics facilities. I assign each facility to its 2020 commuting zone and define a commuting zone’s entry year as the

year its first facility opened.

The main outcomes come from published American Community Survey 1-year estimates for 2005–2024, which the Census Bureau releases for every county with at least 65,000 residents; 2020 is missing because standard 1-year estimates were not released for that year. For adults aged 25 and older, I compute the shares with at least a high school diploma, some college, a bachelor’s degree, and a master’s degree, overall and by gender, and the bachelor’s share by race and ethnicity. I sum the population counts in each category across the counties of each 2020 commuting zone before computing shares. Because the estimates cover only larger counties, the panel describes the more populous part of each commuting zone.

College enrollment data are obtained from the Integrated Postsecondary Education Data System (IPEDS) Fall Enrollment survey for 2005 to 2023. One record per institution and year is retained to ensure each student is counted only once. The primary enrollment outcomes are part-time and full-time enrollment, measured both as the proportion of students enrolled part-time and as the absolute numbers of students enrolled part-time and full-time. Following [Charles et al. \(2018\)](#), I also measure first-time enrollment of degree-seeking undergraduates separately for four-year and two-year institutions, expressed as a percentage of the commuting zone’s population aged 18 to 24. To account for the inclusion of non-local students in college enrollment figures, enrollment at local-draw colleges is constructed by excluding institutions where the share of first-time students from other states or abroad was 30 percent or more in 2006 and 2008. Additionally, IPEDS has reported distance-education enrollment since 2012.

To examine the determinants of changes in attainment and enrollment, I utilize five additional data sources. Data on household income, rents, poverty, unemployment, labor force participation, employment by industry and occupation, hourly wages, and migration status (defined as whether an individual resided in the same state one year earlier) are drawn from the IPUMS American Community Survey (ACS) 1-year microdata. The ACS delineates Public Use Microdata Areas (PUMAs), which I assign to counties using population-weighted allocation factors and a crosswalk that aligns with each year’s PUMA definitions (2000 PUMAs for 2005–2011, 2010 PUMAs for 2012–2021, and 2020 PUMAs for 2022–2024), and subsequently to 2020 commuting zones. Because the allocation factor remains constant within a PUMA, I first aggregate person weights within PUMA-year cells and then allocate these cells, thereby precisely reproducing the person-level allocation. Employment by 2-digit NAICS sector is sourced from County Business Patterns (CBP), 2005–2023; for years prior to 2017, cells withheld for disclosure are imputed as the midpoint of the reported employment size class. Establishment counts by sector are obtained from the Census Bureau’s Business Dynamics Statistics (BDS), 2005–2023. County-to-county migration flows and the income of movers and non-movers are derived from the IRS Statistics of Income migration data, 2011–2023. Employment by county and industry for private employers is taken from the

Census Bureau’s Quarterly Workforce Indicators ([U.S. Census Bureau, 2026](#)), 2005–2023, which are constructed from state unemployment-insurance records. Additionally, I incorporate real-time labor demand data from Lightcast (formerly Emsi Burning Glass) for 2014–2023, as described in [Andreadis et al. \(2025\)](#). Using these data, I calculate the share of job postings related to artificial intelligence and other advanced-technology occupations.

County-level data are aggregated to 2020 commuting zones, so that each unit represents a single local labor market. The resulting panel consists of 297 commuting zones observed annually from 2005 to 2024, with 188 zones receiving an Amazon facility between 2010 and 2024. As an independent check on the attainment results, a second panel of attainment shares is constructed from the IPUMS microdata. This panel covers all 577 commuting zones observed annually, including those without a county of 65,000 residents, of which 204 receive an Amazon facility. The analysis of this panel is presented in Online Appendix Section [OA4](#).

3.2 Empirical Specification

If I compare education levels in places with and without Amazon facilities, the results are likely to be biased. Amazon facilities opened in large, prosperous, and well-educated areas, so a simple comparison would give Amazon credit for differences that were already there. Instead of using this approach, I focus on when Amazon entered each area. I compare how outcomes changed in commuting zones after Amazon arrived to how they changed in zones where Amazon had not yet opened during the same years.

Let G_c denote the year in which commuting zone c is first treated. To allow for anticipation, for example hiring and construction before a facility opens, I set G_c to the year before the first facility opens.⁷ In my data, no commuting zone loses all of its Amazon facilities after entry, so treatment is absorbing. Once Amazon enters, it stays, and the estimates capture the cumulative effect of all facilities that follow the first. The event-study specification is

$$Y_{ct} = \alpha_c + \lambda_t + \sum_{e \neq -1} \theta_e \mathbf{1}[t - G_c = e] + \varepsilon_{ct}, \quad (4)$$

where Y_{ct} is an outcome, such as the share of adults with a bachelor’s degree, in commuting zone c and year t , α_c and λ_t are commuting-zone and year effects, and θ_e is the effect e years after treatment, relative to the year before treatment.

⁷Anticipation effects can occur as local residents and markets react to the announcement of a new FC before it becomes operational. This can include pre-opening hiring for construction and management. Supporting this timeframe, [Chung \(2023\)](#) finds that the average period between an FC’s announcement and the actual start of its operations is approximately six months. A one-year window therefore provides a reasonable period to capture these anticipatory dynamics.

Amazon’s entry is staggered over fifteen years, and its effects grow over time. In this setting, estimating equation (4) with a two-way fixed effects regression uses commuting zones treated earlier as comparisons for those treated later and can assign negative weights to some treatment effects (Goodman-Bacon, 2021; Sun and Abraham, 2021; De Chaisemartin and d’Haultfœuille, 2020; Borusyak et al., 2024; Roth et al., 2023). I instead estimate the coefficients with the estimator of Callaway and Sant’Anna (2021). Its building block is the effect for the cohort of commuting zones first treated in year g , measured in year t ,

$$ATT(g, t) = \mathbb{E} [Y_{ct} - Y_{c, g-1} \mid G_c = g] - \mathbb{E} [Y_{ct} - Y_{c, g-1} \mid G_c > t], \quad (5)$$

the change in the outcome since the year before treatment in cohort g , minus the same change in commuting zones not yet treated in year t . I estimate each $ATT(g, t)$ with the doubly-robust estimator, which equals equation (5) when no covariates are included. The event-study coefficients average these effects across cohorts at each event time,

$$\theta_e = \sum_g \Pr(G_c = g \mid G_c + e \leq T) ATT(g, g + e), \quad (6)$$

and the headline estimate is the simple aggregation of Callaway and Sant’Anna (2021), which averages all post-entry group-time effects and weights each cohort by its size,

$$\theta^S = \sum_g \sum_{t \geq g} w_{g,t} ATT(g, t), \quad w_{g,t} = \frac{\Pr(G_c = g)}{\sum_{g'} \sum_{t' \geq g'} \Pr(G_c = g')}. \quad (7)$$

The event-study figures report θ_e for $e = -6, \dots, 6$.

The primary estimation sample comprises commuting zones that ultimately receive a facility, with the comparison group consisting of locations that Amazon enters at a later stage. This restriction ensures comparisons are made only among areas Amazon deems suitable for facility placement. Time-varying local characteristics, which may be influenced by Amazon’s entry, are excluded as controls. The main estimates do not include covariates. In robustness checks that incorporate controls, the estimator conditions on two baseline characteristics measured prior to any treatment: the logarithm of population in 2005–2009 and the change in log population between 2005–2006 and 2008–2009.

The commuting zones that have not yet received a facility, which Amazon enters at a later date, serve as the counterfactual for how attainment would have evolved in the absence of Amazon’s entry.

3.3 Threats to Identification

The main concern is that the parallel trends assumption might not hold. Early-entry locations could have had different trends in education shares than later-entry locations, especially if Amazon chose areas where education levels were already rising. To address this, I use three methods: balance tests on pre-treatment changes between early and later entrants (Marcus and Sant’Anna, 2021), checking pre-treatment coefficients in event studies, and placebo tests that move the entry date earlier (see Sections 4.1 and 6). Because the estimator looks at changes over time, any differences in starting levels between early and later entrants do not affect the estimates.

Another issue is the possibility of a local shock happening at the same time as Amazon’s entry. Some local changes, like rezoning or public investment to attract or support a facility, are part of Amazon’s arrival and are included in the estimates. This is important for places considering whether to attract a facility. A shock unrelated to Amazon would affect the estimates only if it happened in the same years Amazon entered those areas, which is unlikely since entry was spread over fifteen years and 188 commuting zones. The types of changes I see also suggest Amazon’s influence, as most growth is in professional and technical services and skilled jobs (see Section 5.2).

I use a multiplier bootstrap with 1,000 draws for inference, clustering at the commuting-zone level to account for serial correlation within each zone. The event-study figures show uniform confidence bands that cover all event-time coefficients together.

4 Results

In this section, I share the main results on how Amazon’s entry affected local educational outcomes. I use four different data sources, each helping to fill gaps in geographic coverage, migration, or enrollment, so I can provide a more comprehensive analysis. First, I describe the commuting zones where Amazon opened and check if early and later locations had similar trends before Amazon arrived. I then report the effects on educational attainment, adult population composition, and college enrollment. Section 5 looks at possible reasons behind these results.

4.1 Descriptive Statistics and Balance

Table 1 presents characteristics of commuting zones in the sample during 2005 to 2009, prior to Amazon’s expansion. Amazon primarily entered large, urban, and relatively affluent areas. On average, commuting zones that received a facility had 1.22 million residents, approximately 8 times the population of zones Amazon never entered (Appendix Table A1). The median household income in these zones was \$68,939, compared to \$60,371 in zones without Amazon facilities. Additionally, 27.2 percent of adults in treated zones held a bachelor’s degree, versus 23.4 percent in untreated

zones. The labor market in treated zones was more skilled: college graduates held 27.0 percent of jobs and STEM occupations accounted for 4.6 percent of employment, compared to 21.8 and 3.2 percent, respectively, in untreated zones. Employment in transportation and warehousing, Amazon's sector, was only marginally higher in treated zones (3.7 versus 3.4 percent), indicating that Amazon did not exclusively target areas with pre-existing logistics concentrations. Entry accelerated in later years: 66 percent of treated zones received their first facility in 2018 or later, and by 2024, the average treated zone hosted 6.3 facilities.

These differences in baseline characteristics motivate the comparison between commuting zones that Amazon entered earlier and those it entered later, rather than comparing with zones never entered. For identification, it is essential to determine whether early and later entrants exhibited similar pre-entry trends. Following [Marcus and Sant'Anna \(2021\)](#) and [Cunningham \(2024\)](#), treated commuting zones are grouped by the year of their first facility, and each group is compared to zones entered subsequently, considering only pre-treatment years. Early entrants are larger and more educated in terms of levels (Appendix Table [A3](#)). However, in 40 out of 44 comparisons, annual pre-entry changes do not differ significantly between early and later entrants (Appendix Table [A2](#)). The four exceptions occur in the 2014 to 2016 group, where the shares of adults with a high school diploma and with some college increased more slowly, and the college-graduate share of employment and unemployment rate increased more rapidly, compared to later entrants.

4.2 Educational Attainment

Figure [2](#) presents the event-study estimates for the four educational attainment measures, while Table [2](#) summarizes the aggregated effects. The entry of Amazon increases the proportion of adults with a bachelor's degree by 0.54 percentage points (standard error 0.19), representing approximately 2 percent of the pre-entry mean of 28.0 percent. Similarly, the share of adults with a master's degree rises by 0.42 percentage points (standard error 0.12), or about 4 percent of its mean. In contrast, the estimates for high school diploma attainment and some college are small and statistically insignificant. These effects accumulate over time: the bachelor's degree share remains unchanged in the year of entry, increases by 0.21 percentage points after two years, and reaches 1.3 percentage points after six years. The master's degree share increases to 0.8 percentage points after six years. The event-study coefficients become statistically significant from the fourth year following entry. This upward trend is not attributable to changes in the composition of commuting zones over event times, as it is also observed in a balanced sample of commuting zones tracked for all seven years after entry (Section [6](#)).

The pre-treatment coefficients are approximately zero for all four educational outcomes. The observed gains are broad-based, exhibiting similar magnitudes for both men and women, and are

Table 1: Education, Enrollment, and Labor Markets Before Amazon’s Expansion

	All commuting zones (1)	Receive Amazon facility (2)	Never receive facility (3)	Difference (2) – (3) (4)
<i>Panel A. Educational attainment, adults aged 25+ (%)</i>				
High school diploma or more	85.9 (4.8)	86.2 (4.4)	85.5 (5.3)	0.7 [0.6]
Some college or more	55.3 (7.8)	56.5 (6.9)	53.0 (8.8)	3.5*** [1.0]
Bachelor’s degree or more	25.8 (7.5)	27.2 (6.6)	23.4 (8.3)	3.9*** [0.9]
Master’s degree or more	9.4 (3.6)	9.8 (3.2)	8.5 (4.0)	1.3*** [0.4]
<i>Panel B. College enrollment (IPEDS)</i>				
First-time students, four-year colleges (% of ages 18–24)	3.84 (3.36)	4.11 (3.12)	3.38 (3.72)	0.73* [0.42]
First-time students, two-year colleges (% of ages 18–24)	4.25 (2.23)	4.20 (1.84)	4.33 (2.79)	–0.13 [0.30]
Part-time students (% of enrollment)	36.9 (12.1)	36.0 (10.9)	38.4 (13.8)	–2.4 [1.5]
<i>Panel C. Income and labor market</i>				
Median household income (2023 \$)	65,795 (12,084)	68,939 (12,465)	60,371 (9,179)	8,569*** [1,265]
Poverty rate (%)	13.1 (3.9)	12.5 (3.6)	14.1 (4.1)	–1.6*** [0.5]
Unemployment rate (%)	7.1 (1.5)	7.0 (1.4)	7.2 (1.8)	–0.2 [0.2]
Labor force participation rate (%)	64.1 (4.5)	65.2 (4.0)	62.4 (4.8)	2.8*** [0.5]
College graduates (% of employment)	25.1 (6.4)	27.0 (6.2)	21.8 (5.3)	5.3*** [0.7]
STEM occupations (% of employment)	4.1 (1.8)	4.6 (1.8)	3.2 (1.2)	1.4*** [0.2]
Abstract-task occupations (% of employment)	32.2 (4.1)	33.4 (4.0)	30.2 (3.5)	3.2*** [0.4]
Professional & technical services (% of employment)	4.8 (2.4)	5.5 (2.5)	3.7 (1.5)	1.7*** [0.2]
Transportation & warehousing (% of employment)	3.6 (1.5)	3.7 (1.5)	3.4 (1.5)	0.3* [0.2]
Commuting zones	297	188	109	

Notes: Each commuting zone’s characteristics are averaged over 2005–2009, before any commuting zone in the sample is treated, and the table reports the mean across commuting zones, with standard deviations in parentheses; commuting zones are not weighted by population. Column (2) contains commuting zones where Amazon opened at least one facility between 2010 and 2024, column (3) the remaining commuting zones in the balanced 2005–2024 panel. Column (4) reports the difference in means, with its standard error in brackets; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Attainment is from published ACS 1-year estimates for counties with at least 65,000 residents, aggregated to 2020 commuting zones; income, poverty, labor-market, and occupation measures are from IPUMS ACS microdata for the same commuting zones; household income is converted to 2023 dollars with the CPI-U. Enrollment is from IPEDS (first-time students, 2006–2009): first-time degree-seeking undergraduates at local-draw colleges as a percentage of the population aged 18–24, and part-time students as a percentage of all students (commuting zones with colleges). STEM occupations combine computer and mathematical, engineering, and science occupations; abstract-task occupations are the top employment-weighted third of the Autor and Dorn (2013) abstract task index. Industry shares are from County Business Patterns.

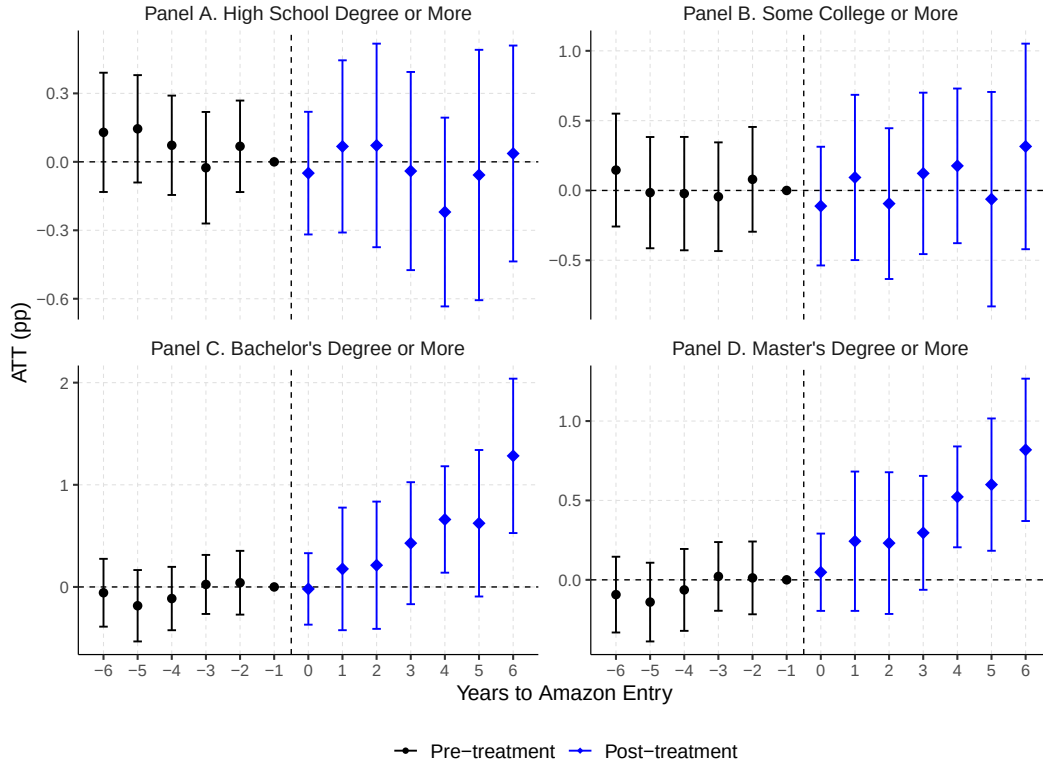
precisely estimated for White adults. However, estimates for smaller demographic groups remain imprecise (Appendix Table A4).

4.3 Migration and the Composition of the Adult Population

An increased share of adults with a degree may result from either a greater number of residents earning degrees or an influx of degree holders. Table 3 analyzes both possibilities using the same specification as the attainment results.

Panel A demonstrates that the increase in the share of adults with a degree is attributable to

Figure 2: Event-Study Estimates: Effects of Amazon’s Entry on Educational Attainment, 2005–2024



Notes: Each panel plots event-time coefficients; black circles are pre-treatment and blue diamonds post-treatment coefficients. Bars show uniform (simultaneous) 95% confidence bands from the multiplier bootstrap, clustered at the commuting-zone level. Event time $t = -1$ is normalized to zero and the vertical dashed line marks the treatment year, defined as the year before Amazon’s first facility opens in the commuting zone (to allow for anticipation). Dependent variable is the percentage-point share of adults aged 25 and older completing at least the specified schooling level, from published American Community Survey 1-year estimates for counties with at least 65,000 residents (2005–2024; 2020 is not available), aggregated to 2020 commuting zones. Sample is the balanced panel of commuting zones that receive an Amazon facility between 2010 and 2024. Estimates are the doubly-robust staggered difference-in-differences event-study coefficients of [Callaway and Sant’Anna \(2021\)](#), comparing treated commuting zones to not-yet-treated controls.

growth in the number of graduates. According to published ACS estimates, the number of adults with a bachelor’s degree increases by 2.6 percent, approximately twice the rate of those without a degree, who increase by 1.4 percent. Place of birth, a fixed characteristic, indicates whether the increase is concentrated among individuals originating from outside the state. The number of graduates born in their state of residence increases by 2.4 percent, while those born in another state increase by 2.2 percent, indicating that graduates from other states do not grow faster than local graduates. In contrast, the number of foreign-born graduates increases more rapidly, by 6.6 percent.

Panel B examines recent migration patterns, indicating that Amazon’s entry attracts educated individuals. While the total number of adults who moved from another state or from abroad in

Table 2: Effects of Amazon’s Entry on Educational Attainment

	High School+ (1)	Some College+ (2)	Bachelor’s+ (3)	Master’s+ (4)
Amazon Entry	−0.013 (0.138)	0.055 (0.167)	0.541*** (0.185)	0.422*** (0.117)
Mean Dep. Var.	87.54	58.46	27.99	10.29
SD	4.59	7.14	6.60	3.28
Commuting zones	188	188	188	188
N	3,572	3,572	3,572	3,572

Notes: Dependent variable is the percentage-point share of adults aged 25 and older completing at least the specified schooling level, from published American Community Survey 1-year estimates for counties with at least 65,000 residents (2005–2024; 2020 is not available), aggregated to 2020 commuting zones. Estimates are the doubly-robust staggered difference-in-differences estimator of [Callaway and Sant’Anna \(2021\)](#), aggregated over all post-entry years (simple ATT), comparing treated commuting zones to not-yet-treated controls. Sample is the balanced panel of commuting zones that receive an Amazon facility between 2010 and 2024; a commuting zone is treated from the year before its first facility opens (anticipation). Standard errors from the multiplier bootstrap (1,000 draws), clustered at the commuting-zone level, in parentheses. Mean Dep. Var. and SD report the pre-treatment mean and standard deviation in treated commuting zones. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

the past year remains unchanged, the proportion of graduates among these movers increases.⁸ The number of recent in-migrants with a bachelor’s degree increases by 0.068 per 100 adults, representing approximately 7 percent of their pre-entry level, and those with a master’s degree increase by 0.041, or 11 percent. In contrast, the number of recent in-migrants without a degree does not change. None of these series shows significant differences prior to entry. The inflow of graduates is substantial relative to the overall change in the stock. If all additional graduates who moved in remained, the cumulative inflow would average 0.35 graduates per 100 adults over the post-entry years, compared to the 0.54 percentage point increase in the bachelor’s share in the main estimates. Therefore, in-migration from other states and abroad can explain at most approximately two-thirds of the increase. Supporting this, the bachelor’s share increases by 0.80 percentage points among adults who did not move in the past year.

Panel C utilizes IRS records, which capture all movers but do not provide information on educational attainment. The number of individuals moving into the area remains unchanged, while the number moving out increases slightly (1.02 per 1,000 residents, significant at the 10 percent level). Household incomes increase by similar percentages for both incoming movers (3.1 percent) and households that remain (3.0 percent). Consequently, the income data do not provide additional evidence regarding the educational composition of movers.

Newcomers from other states and from abroad account for at most about two-thirds of the

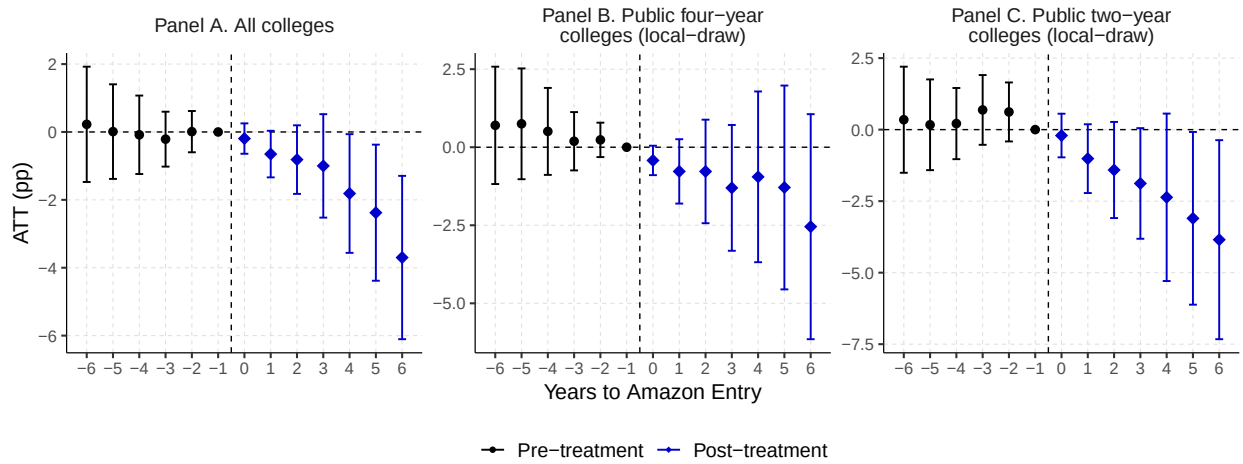
⁸Graduates are more likely than non-degree holders to relocate in response to local demand shifts and tend to move to areas with rising demand for skilled labor ([Bound and Holzer, 2000](#); [Diamond, 2016](#)).

increase in the share of adults with a bachelor’s degree. The remaining third or more comes from people already living in the area, either residents who complete degrees or graduates who stay instead of leaving. Because the data do not follow the same people over time or record moves within a state, I cannot separate these two groups. The local share is consistent with upskilling, although some of it may reflect graduates who would otherwise have left.

4.4 College Enrollment

Attainment begins to rise within two years of Amazon’s entry, a period too brief for these changes to result primarily from new students completing four-year degrees. These early gains likely arise from two alternative mechanisms. The first is migration, as indicated in Section 4.3, where part of the increase may be attributed to graduates remaining in or relocating to commuting zones where Amazon has established a presence. The second mechanism is accelerated degree completion among students already enrolled, as those who transition from part-time to full-time study can graduate more quickly. Enrollment data provide suggestive evidence supporting this latter response.

Figure 3: Event-Study Estimates: Part-Time Share of Enrollment, 2010–2023



Notes: Each panel plots event-time coefficients for part-time enrollment as a percentage of total enrollment at colleges of the specified type in the commuting zone. The sample is restricted to commuting zones that receive an Amazon facility, with not-yet-treated commuting zones as comparisons (Table 4). Black circles are pre-treatment and blue diamonds post-treatment coefficients. Local-draw colleges exclude institutions whose share of first-time students from other states or abroad was 30 percent or more in 2006 and 2008. Bars show uniform (simultaneous) 95% confidence bands from the multiplier bootstrap, clustered at the commuting-zone level. Event time $t = -1$ is normalized to zero and the vertical dashed line marks the treatment year, defined as the year before Amazon’s first facility opens in the commuting zone. Enrollment is from IPEDS Fall Enrollment, 2010–2023. Estimates are the doubly-robust staggered difference-in-differences event-study coefficients of Callaway and Sant’Anna (2021).

Amazon’s entry alters student attendance patterns. In commuting zones that receive a facility, the proportion of students enrolled part time decreases by 1.62 percentage points, with the decline

Table 3: Amazon’s Entry, Migration, and the Composition of the Adult Population

	Pre-entry mean	Effect of Amazon’s entry	Commuting zones
<i>Panel A. Adults by degree and place of birth, published ACS estimates (log, effect in percent)</i>			
Adults without a bachelor’s degree	439.0	1.35** (0.65)	188
Adults with a bachelor’s degree	191.5	2.62*** (0.91)	188
born in state of residence	78.4	2.42** (1.21)	188
born in another state	79.5	2.23* (1.24)	188
foreign-born	30.7	6.62** (2.71)	188
<i>Panel B. Adults who moved from another state or from abroad in the past year, per 100 adults, ACS microdata</i>			
All	2.69	0.086 (0.059)	204
Without a bachelor’s degree	1.75	0.018 (0.045)	204
With a bachelor’s degree	0.94	0.068** (0.031)	204
With a master’s degree	0.37	0.041*** (0.015)	204
Bachelor’s share, adults who did not move (pp)	25.07	0.80*** (0.17)	204
<i>Panel C. Migration flows and household income, IRS records (2011–2023)</i>			
In-migrants per 1,000 residents	48.15	0.19 (0.48)	195
Out-migrants per 1,000 residents	48.00	1.02* (0.52)	197
Income of in-migrant households	32.9	3.06** (1.21)	195
Income of households that did not move	43.5	3.00*** (0.66)	203
In-migrant income, % of non-movers’	76.25	0.23 (0.92)	195

Notes: Each row reports the simple ATT of Amazon’s entry, which averages all post-entry group-time effects, estimated as in Table 2: the doubly-robust estimator of Callaway and Sant’Anna (2021) with not-yet-treated comparison commuting zones, no covariates, and treatment in the year before the first facility opens. Panel A uses the published ACS 1-year county estimates of the main analysis: counts of adults aged 25 and older by degree (table B15002) and of adults with a bachelor’s degree by place of birth (table B06009). Pre-entry means of these counts are in thousands of persons, and effects are in percent (100 times log points). Panel B uses IPUMS ACS microdata, which cover all commuting zones. Its first four rows count adults aged 25 and older who lived in another state or abroad one year earlier, per 100 adults aged 25 and older; adults who did not move lived in the same state one year earlier. Panel C uses IRS Statistics of Income county-to-county migration data for 2011–2023; migrants are counted as exemptions, income is adjusted gross income per return in thousands of 1999 dollars (pre-entry means in levels, effects in percent), and commuting zones treated before 2012 have no pre-entry year and drop out. Standard errors from the multiplier bootstrap (1,000 draws), clustered at the commuting-zone level, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Part-Time Share of Enrollment, 2010–2023

	All colleges (1)	Public four-year (local-draw) (2)	Public two-year (local-draw) (3)
Amazon Entry	–1.624*** (0.564)	–1.179 (0.726)	–2.068*** (0.641)
Commuting zones	192	159	170
Mean Dep. Var.	36.40	28.76	56.22

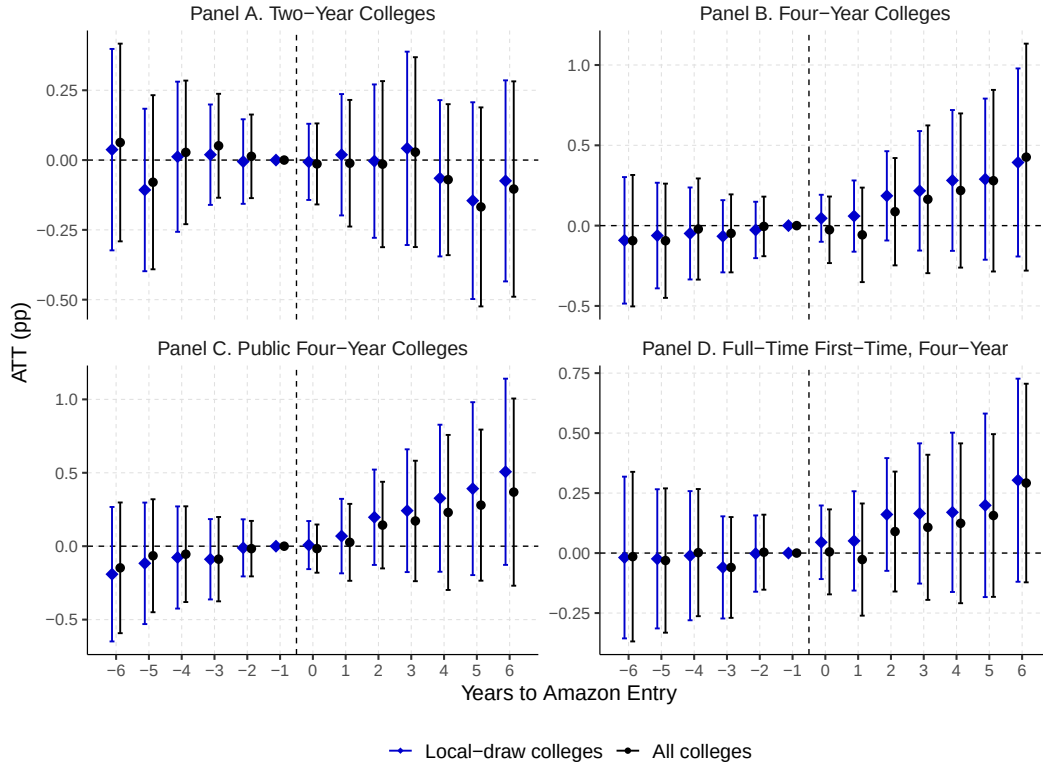
Notes: Dependent variable is part-time enrollment as a percentage of total enrollment (all students, undergraduate and graduate) at colleges of the specified type located in the commuting zone, 2010–2023, from IPEDS Fall Enrollment. Four-year colleges are bachelor’s-granting institutions and two-year colleges do not award bachelor’s degrees. Local-draw colleges exclude institutions whose share of first-time students from other states or abroad was 30 percent or more in 2006 and 2008. A commuting zone is treated from the year before its first Amazon facility opens; cohorts treated in 2010 have no pre-period and do not contribute. The sample is restricted to commuting zones that receive an Amazon facility, so commuting zones treated later serve as comparisons for those treated earlier. Estimates are the doubly-robust staggered difference-in-differences ATT of [Callaway and Sant’Anna \(2021\)](#), aggregated over all post-entry years (simple ATT); the estimator differences out time-invariant commuting-zone characteristics and common year shocks. Standard errors from the multiplier bootstrap (1,000 draws), clustered at the commuting-zone level, in parentheses. Mean Dep. Var. is the pre-treatment mean in treated commuting zones. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

expanding to 3.7 percentage points after six years (Table 4; Figure 3). This reduction reflects two trends: an increase in full-time enrollment at public four-year colleges and a decrease in part-time enrollment at public two-year colleges, consistent with the substitution effect described in the framework for students balancing work and study. Prior to the pandemic, the proportion of students engaged in fully online study at local public colleges also declined (Online Appendix Table OA4).

The shift toward full-time study is also evident among new students. New enrollment is measured following [Charles et al. \(2018\)](#) as the proportion of first-time, degree-seeking undergraduates at four-year and two-year colleges in the commuting zone, relative to the population aged 18 to 24. Because IPEDS records enrollment by college location, the analysis focuses on local-draw colleges, excluding institutions that enrolled 30 percent or more of their first-time students from other states or abroad in 2006 and 2008. This exclusion removes 34 percent of four-year and 3 percent of two-year first-time enrollment. First-time enrollment at local four-year colleges increases by 0.262 percentage points (standard error 0.134), representing approximately 5 percent of the pre-entry mean of 5.08. The effect at public four-year colleges rises to 0.51 percentage points six years after entry (Appendix Table A5; Figure 4). The increase is concentrated among full-time students (0.199, standard error 0.100). Enrollment at two-year colleges remains unchanged (–0.057). These estimates are less precise than the attainment results and are significant at the 10 percent level.

Enrollment appears to shift toward full-time, in-person study at four-year colleges rather than out of the education system entirely. Because IPEDS reports enrollment by college and year without

Figure 4: Event-Study Estimates: First-Time College Enrollment, 2010–2023



Notes: Each panel plots event-time coefficients for first-time enrollment as a percentage of the commuting zone’s population aged 18–24, estimated separately for local-draw colleges (blue diamonds), which exclude institutions whose share of first-time students from other states or abroad was 30 percent or more in 2006 and 2008, and for all colleges (black circles). Panels A–C use first-time degree- or certificate-seeking undergraduate enrollment; Panel D uses full-time first-time enrollment at four-year colleges. Two-year colleges do not award bachelor’s degrees; four-year colleges are bachelor’s-granting institutions (Charles et al., 2018). Bars show uniform (simultaneous) 95% confidence bands from the multiplier bootstrap, clustered at the commuting-zone level. Event time $t = -1$ is normalized to zero and the vertical dashed line marks the treatment year, defined as the year before Amazon’s first facility opens in the commuting zone. Enrollment is from IPEDS Fall Enrollment (2010–2023) aggregated to 2020 commuting zones; population is from IPUMS ACS 1-year microdata. Estimates are the doubly-robust staggered difference-in-differences event-study coefficients of Callaway and Sant’Anna (2021), comparing treated commuting zones with not-yet-treated.

tracking individual students, it is not possible to determine whether the same students transition from part-time to full-time study; therefore, this evidence should be considered suggestive. As full-time students typically complete degrees more rapidly, this shift may also contribute to increased attainment.

5 Mechanisms

The conceptual framework identifies two mechanisms through which Amazon’s entry may increase educational attainment, despite the substitution effect of its warehouse employment. First, higher household income relaxes financial constraints. Second, increased demand for skilled labor at other firms, resulting from Amazon’s spillover effects, raises the expected return to obtaining a degree.

5.1 Household Income

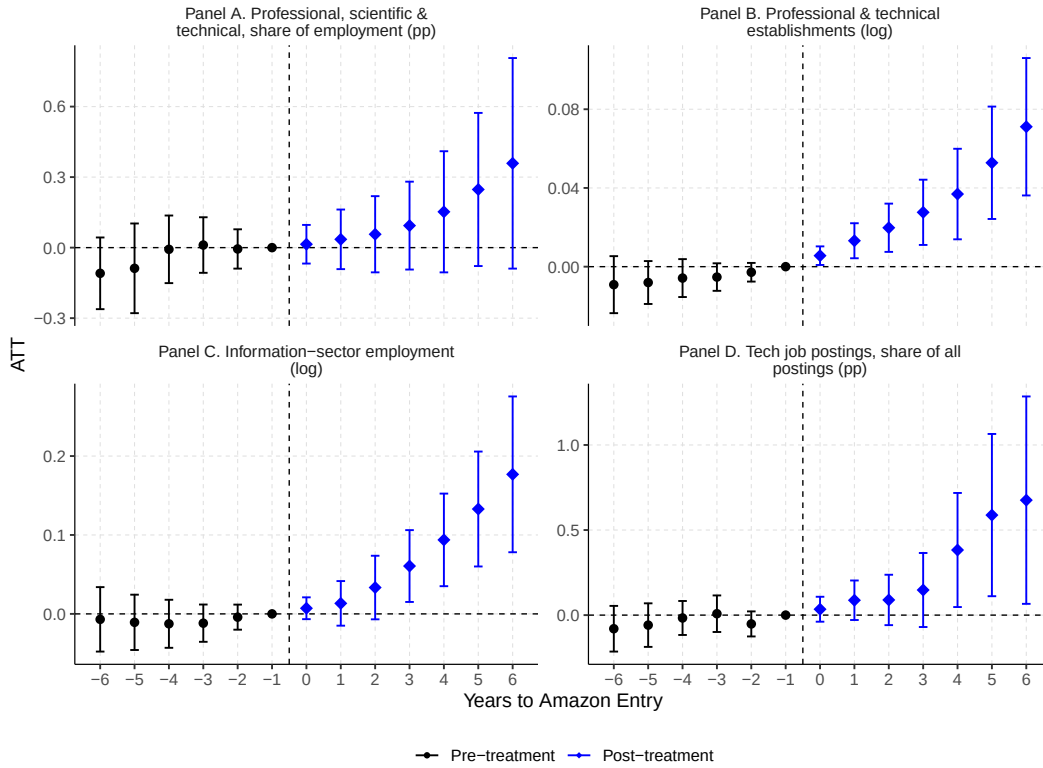
Median household income increases gradually following Amazon’s entry, rising by 1.5 percent on average, or approximately \$960 per household in 2023 dollars, and by 3.9 percent six years after entry (Appendix Table A6, Panel A). The mean wage of workers remains unchanged, while the unemployment rate declines modestly. The poverty rate also remains unchanged. Rents increase more rapidly than income, by 4.0 percent on average. If the observed education effects were attributable to relaxed credit constraints, the largest effects would be expected in areas with the highest pre-existing poverty rates. However, this pattern is not observed. When treated commuting zones are divided at the median of the 2005–2009 poverty rate, household income increases by 3.1 percent in lower-poverty commuting zones and does not increase in higher-poverty zones. The effect on bachelor’s degree attainment is also greater in the lower-poverty half (0.60 versus 0.25 percentage points; Appendix Table A6, Panel B). Therefore, the income channel is unlikely to account for the observed attainment results.

5.2 The Demand for Skilled Work

Amazon’s facilities primarily function as warehouses, and the majority of positions within the company do not require a college degree. However, Figures 5 and 6 and Table 5 demonstrate that Amazon’s entry into a region shifts local employment toward skilled occupations, with much of this transition occurring at other firms.

The share of employment in professional, scientific, and technical services—including engineering, consulting, accounting, and computer services—increases by 0.18 percentage points on average and by 0.36 percentage points six years after Amazon’s entry. The number of establishments in these sectors grows by 3.9 percent, with significant increases observed in every year following entry (Figure 5 and Appendix Table A9). Administrative records from the Quarterly Workforce Indicators corroborate this trend, indicating that employment in the information sector rises by 8.8 percent after Amazon’s arrival. Job postings data further support these findings: the share of postings for artificial intelligence and other advanced-technology jobs increases by 0.18 percentage points from a

Figure 5: Event-Study Estimates: Skilled Demand at Other Firms



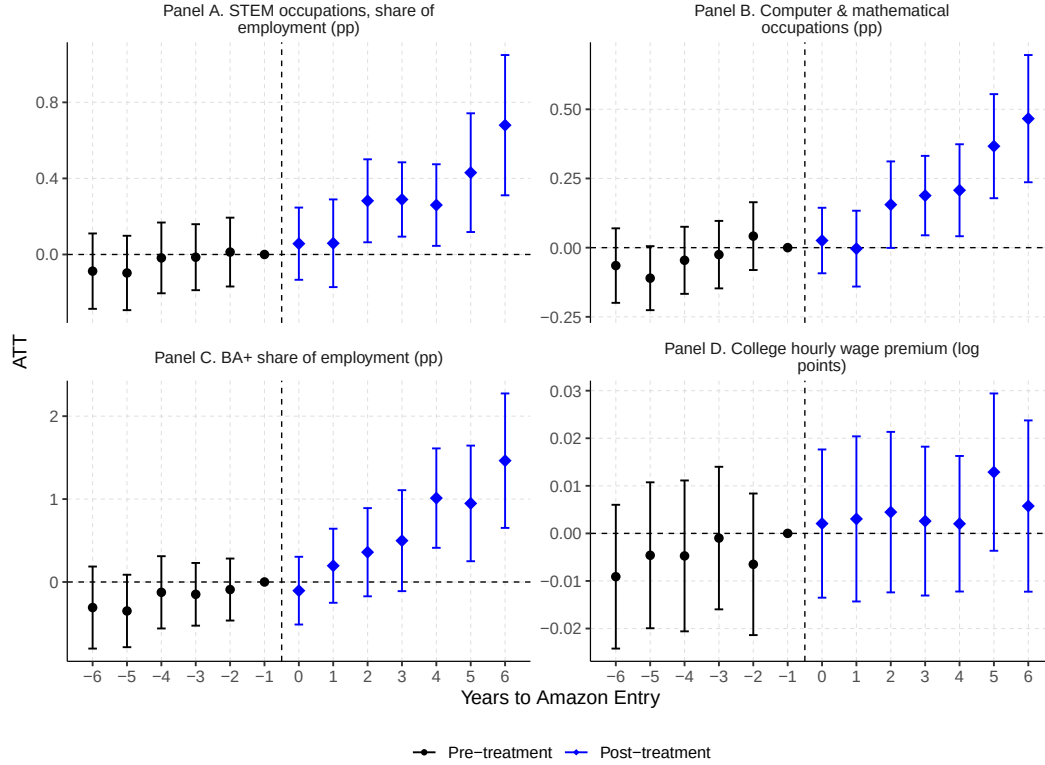
Notes: Each panel plots event-time coefficients from the estimator of [Callaway and Sant'Anna \(2021\)](#) with not-yet-treated comparisons in commuting zones that receive an Amazon facility; black circles are pre-treatment and blue diamonds post-treatment coefficients, with uniform 95% confidence bands from the multiplier bootstrap (1,000 draws), clustered at the commuting-zone level. Panel A: employment in professional, scientific, and technical services (NAICS 54) as a percentage of total employment, County Business Patterns, 2005–2023. Panel B: log number of establishments in NAICS 54, Business Dynamics Statistics, 2005–2023. Panel C: log employment in the information sector (NAICS 51), private employers, Quarterly Workforce Indicators ([U.S. Census Bureau, 2026](#)), 2005–2023. Panel D: postings for artificial intelligence and other advanced-technology jobs as a percentage of all job postings, Lightcast, 2014–2023, from the county-level data of [Andreadis et al. \(2025\)](#); commuting zones first treated by 2014 have no pre-entry year in this window and are excluded. Treatment begins the year before Amazon's first facility opens.

pre-entry mean of 0.47 percent, and by approximately 0.7 percentage points six years after entry.⁹ In contrast, the shares of employment in retail trade, accommodation and food services, administrative support, and manufacturing decline.

Amazon's entry also alters the composition of the workforce. The share of employment in STEM occupations increases by 0.34 percentage points from a base of 4.4 percent, primarily due to growth in computer and mathematical occupations, which rise by approximately 12 percent relative to their

⁹Job-posting counts are from Lightcast (formerly Emsi Burning Glass) at the county level for 2014–2023, from the county-level data of [Andreadis et al. \(2025\)](#). Online job postings are widely used to measure changes in local skill demand ([Hershbein and Kahn, 2018](#)). Technology postings are postings for artificial intelligence and other advanced-technology jobs as classified in those data, and the outcome is their share of all postings. Commuting zones Amazon entered by 2014 have no pre-entry year in this window, which leaves 145 commuting zones.

Figure 6: Event-Study Estimates: Skilled Jobs and the College Wage Premium



Notes: Each panel plots event-time coefficients from the estimator of [Callaway and Sant'Anna \(2021\)](#) with not-yet-treated comparisons in commuting zones that receive an Amazon facility; black circles are pre-treatment and blue diamonds post-treatment coefficients, with uniform 95% confidence bands from the multiplier bootstrap (1,000 draws), clustered at the commuting-zone level. Outcomes are computed from IPUMS American Community Survey 1-year microdata, 2005–2024: the shares of employment in STEM and in computer and mathematical occupations, the share of employment held by workers with at least a bachelor's degree, and the college hourly wage premium (the difference in mean log hourly wages between workers with and without a bachelor's degree). Treatment begins the year before Amazon's first facility opens.

pre-entry level. Professional and managerial occupations gain 0.46 percentage points, while the share of employment in abstract-task occupations, as defined by [Autor and Dorn \(2013\)](#), increases by 0.48 percentage points. The share of clerical, sales, and office jobs changes by -0.52 percentage points, and transportation and material moving occupations, including warehouse work, rise by only 0.16 percentage points. Additionally, the proportion of jobs held by college graduates increases by 0.74 percentage points (Figure 6).

In contrast, the college wage premium remains unchanged. Hourly wages for college graduates (-0.001 log points) and for other workers (-0.008) do not change, nor does the difference between them (0.007, standard error 0.005). According to equation (3), Amazon's entry increases the expected return to a degree by expanding the availability of skilled jobs (p_B), rather than by increasing their pay (Δw_B). The stability of the wage premium, despite rising skilled employment, suggests that the supply of graduates keeps pace with demand, either through retention and in-migration of graduates

Table 5: Effects of Amazon’s Entry on the Demand for College-Educated Workers

	Amazon Entry (1)	Std. error (2)	Mean before entry (3)
<i>Panel A. Education of the workforce</i>			
BA+ share of employment (pp)	0.736***	(0.187)	27.24
<i>Panel B. Occupation share of employment (pp)</i>			
STEM occupations	0.340***	(0.067)	4.39
Computer & mathematical	0.230***	(0.048)	1.93
Professional & managerial	0.464**	(0.182)	33.29
Management, business & financial	0.277***	(0.104)	13.06
Abstract-task occupations (Autor–Dorn)	0.476***	(0.162)	33.17
Sales & office	−0.516***	(0.125)	23.95
Transportation & material moving	0.156*	(0.090)	6.78
<i>Panel C. Wages (log points)</i>			
Log hourly wage, BA+ workers	−0.001	(0.005)	2.87
Log hourly wage, non-BA workers	−0.008**	(0.003)	2.39
College hourly wage premium	0.007	(0.005)	0.47
College annual wage premium	0.001	(0.006)	0.59

Notes: Outcomes are computed from IPUMS ACS 1-year microdata aggregated to 2020 commuting zones, 2005–2024. Panel A: bachelor’s-degree holders as a share of all employed persons aged 16 and older. Panel B: occupation groups of employed persons aged 16 and older, based on the harmonized 2010 occupation codes; STEM combines computer and mathematical, architecture and engineering, and life, physical, and social science occupations; professional and managerial adds management, business, and financial and other professional occupations; abstract-task occupations are those in the top employment-weighted third of the abstract task-intensity index of [Autor and Dorn \(2013\)](#) (occupations mapped to occ1990dd with the crosswalk of David Dorn; thirds set on national employment in 2005–2009). Panel C: hourly wages of workers aged 25–64 are annual wage and salary income divided by weeks worked (interval midpoints) times usual weekly hours, in constant 1999 dollars; the premium is the difference in mean log hourly wages between workers with and without a bachelor’s degree; the annual premium uses the log of mean annual earnings. A commuting zone is treated from the year before its first Amazon facility opens. Each row is a separate doubly-robust staggered difference-in-differences estimate ([Callaway and Sant’Anna, 2021](#)) on the balanced panel of commuting zones that receive an Amazon facility, with not-yet-treated commuting zones as comparisons; column (1) reports the simple ATT, which averages all post-entry group-time effects, and column (2) reports standard errors from the multiplier bootstrap (1,000 draws), clustered at the commuting-zone level. Column (3) is the pre-treatment mean in treated commuting zones (for log outcomes, the geometric mean count). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

or through residents pursuing full-time, four-year degrees.¹⁰ Although the occupation data do not distinguish between skilled jobs at Amazon and those at other employers, the observed growth in professional and technical establishments, information-sector employment, and technology job postings indicates that these changes extend beyond Amazon’s own operations.

¹⁰Amazon’s Career Choice program prepays tuition for its hourly employees ([Amazon, 2021](#)). Before 2021 it covered mainly certificates and associate degrees, and it was extended to bachelor’s degrees only in 2021, late in the sample, so it is unlikely to explain the rise in the shares of adults with a bachelor’s or master’s degree.

6 Robustness and Falsification

In the present section, I show that my key finding is robust over a number of further specifications. Table 6 contains four robustness checks which examine whether the attainment effects could arise even in the absence of Amazon, while Appendix Figure A4 provides a fifth check based on a balanced event window. Panel A uses the main panel for an alternative estimator and a placebo test. Panel B employs the IPUMS microdata panel, which covers all commuting zones, to carry out the inference and the pandemic-related tests.

Table 6: Robustness and Falsification Tests

	Bachelor's degree or more		Master's degree or more	
	No controls (1)	Baseline controls (2)	No controls (3)	Baseline controls (4)
<i>Panel A. Main panel: published ACS estimates, 188 treated commuting zones</i>				
Callaway–Sant’Anna (Table 2)	0.541*** (0.185)		0.422*** (0.117)	
Synthetic difference-in-differences	0.675*** (0.168)		0.446*** (0.113)	
log adult population projected out	0.566*** (0.176)		0.397*** (0.104)	
Placebo: entry moved five years earlier	0.102 (0.102)	0.138 (0.180)	0.051 (0.069)	0.060 (0.095)
<i>Panel B. IPUMS microdata panel, 204 treated commuting zones</i>				
Estimate, 2005–2024	0.826*** (0.161)	0.435** (0.213)	0.605*** (0.082)	0.334*** (0.110)
Randomization-inference p -value	<0.002	0.002	<0.002	<0.002
Pre-pandemic sample, 2005–2019	0.617*** (0.148)	0.428*** (0.166)	0.425*** (0.085)	0.248** (0.099)

Notes: Each cell reports the simple ATT of Amazon’s entry, which averages all post-entry group-time effects, in percentage points of adults aged 25 and older. Panel A uses the published American Community Survey 1-year county estimates of the main analysis (2005–2024; 2020 is not available). Its first row repeats the main estimate from the doubly-robust estimator of Callaway and Sant’Anna (2021) with not-yet-treated comparisons. The synthetic difference-in-differences estimator of Arkhangelsky et al. (2021) uses the 109 commuting zones that never receive a facility as donors, without covariates or with the log adult population projected out; its standard errors come from 200 bootstrap replications. Panel B uses the IPUMS ACS microdata panel, which covers all commuting zones, and the estimator of Callaway and Sant’Anna (2021) with not-yet-treated comparisons. Baseline controls are log population in 2005–2009 and the change in log population between 2005–2006 and 2008–2009. The randomization-inference p -value is the share of 500 random reassignments of treatment years across treated commuting zones, holding the number of commuting zones treated in each year fixed, that produce an estimate at least as large in absolute value as the actual one. The pre-pandemic sample ends in 2019, and commuting zones treated after 2019 serve as comparisons. The placebo, on the main panel, moves each commuting zone’s treatment date five years earlier and uses only years before the actual treatment (172 commuting zones whose fake treatment year is 2007 or later). Standard errors from the multiplier bootstrap (1,000 draws), clustered at the commuting-zone level, in parentheses, unless noted otherwise. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Using an alternative estimator, the results are in line with those from the primary analysis. The main estimates relate the commuting zones that Amazon entered at an early date to those it entered at a later date. By contrast, the synthetic difference-in-differences estimator proposed by [Arkhangelsky et al. \(2021\)](#) compares the zones that Amazon did enter with a weighted combination of zones that it never entered, matching them on pre-entry trends. On the basis of the main panel, this approach gives effects of 0.67 percentage points for the bachelor's share and 0.45 for the master's share, and 0.57 and 0.40 when the log adult population is projected out. These findings are very close to the main estimates of 0.54 and 0.42, and the pre-entry coefficients stay close to zero (Appendix Figure [A3](#)). The synthetic estimator also identifies a small decrease in the share of people who have a high school diploma (−0.25 percentage points; Online Appendix Section [OA4](#)), a feature which is not found in the main estimates.

The effects are probably not the result of random variation. In the microdata panel, the estimated effect is 0.83 percentage points for the bachelor's share and 0.60 for the master's share, and becomes 0.43 and 0.33 when allowing for the baseline population size and growth. In order to evaluate the probability of obtaining spurious results, I randomly reassigned Amazon's entry years among the treated commuting zones 500 times, keeping the number of zones entered each year the same, and reestimated the effect for each of these iterations. The resulting placebo estimates are centered on zero (Appendix Figure [A2](#)), and in no more than one of the 500 random assignments does the estimated effect equal or exceed the observed effect for any outcome or specification (randomization-inference $p \leq 0.002$).

In order to examine existing trends, I carry out a placebo test by advancing each commuting zone's entry by five years and then estimate the impact of this artificial entry on the main panel by using only the years before Amazon's involvement. The placebo effects obtained are very small and statistically not significant: 0.10 percentage points for the proportion with a bachelor's degree ($p=0.32$) and 0.05 for that with a master's degree ($p=0.46$); the values are 0.14 and 0.06 respectively when the baseline controls are included ($p=0.44$ and $p=0.53$).

The effects seen cannot be attributed to changes in the make-up of the commuting zones when the different event times are compared. Since Amazon joined most of the commuting zones towards the later part of the study period, the later event times are mainly based on the early entrants, and this might cause the trends identified in the event study to be confounded by differences between the early and late entrants.¹¹ In order to deal with this issue, I re-estimate the event studies using a balanced event window, keeping only those commuting zones that were observed for all seven years after entry (93 in the case of attainment). The proportion of bachelor's degrees rises steadily, going from 0.20 percentage points in the year of entry to 1.28 percentage points six years later. The

¹¹For a discussion of how changes in cohort composition across event times affect dynamic estimates, see [Callaway and Sant'Anna \(2021\)](#) and [Sun and Abraham \(2021\)](#).

average effect over this period is 0.62 percentage points for the bachelor’s share and 0.42 percentage points for the master’s share (Appendix Figure A4).

Restricting the sample to end in 2019, thereby excluding pandemic years and using commuting zones entered after 2019 as comparisons, yields somewhat smaller estimates that remain positive and significant: 0.62 percentage points for the bachelor’s share and 0.43 for the master’s share without controls, and 0.43 and 0.25 with controls (Table 6, Panel B).

7 Conclusion

Previous research has established that booms in low-skill employment often reduce school attendance among young people. This study provides a comprehensive analysis of the impact of Amazon’s national expansion of logistics facilities on human capital formation, with particular attention to positive spillover effects on other firms. In commuting zones where Amazon enters, the proportion of adults holding a bachelor’s degree increases by 0.54 percentage points, and the share with a master’s degree rises by 0.42 percentage points; both effects intensify over time. These gains likely reflect, in part, graduates who remain in or relocate to these areas, and may also reflect students who transition from part-time to full-time study. The findings are robust to randomization inference, placebo tests with earlier entry dates, a balanced event window, a pre-pandemic sample, and a synthetic difference-in-differences estimator.

Increased demand for skilled labor at other firms appears to drive these effects. Following Amazon’s entry, professional, scientific, and technical establishments expand; information-sector employment and the share of technology job postings increase; and employment shifts toward STEM, managerial, and abstract-task occupations. The college wage premium remains unchanged. Amazon’s spillovers generate additional skilled jobs locally without increasing wages, prompting responses from residents and graduates. The number of degree holders in the area rises, and full-time enrollment in four-year programs increases. Household income grows only modestly, mainly in already affluent areas, suggesting that relaxed financial constraints are an unlikely explanation.

These findings contribute to the ongoing debate regarding the incentives that states and localities provide to attract large employers. Traditional evaluations of such incentives focus primarily on job creation and wage growth. The present analysis indicates that the skill composition of jobs introduced by an employer, as well as subsequent economic activity, also influences the local population’s educational attainment. Several limitations should be noted. The estimates pertain to commuting zones rather than individuals, and individual data linking workers’ education histories to their employers and locations would enable more precise differentiation between new degrees and graduate migration. Additionally, the estimates reflect the overall impact of Amazon’s arrival, including associated local investments, and do not isolate the effect of the logistics facilities alone.

Whether similar effects are produced by other large logistics or technology employers, and whether these effects scale with the magnitude of local incentives, remain open questions.

Declaration on the Use of Generative AI

I used Claude (Anthropic) to assist with language editing and to improve the clarity and readability of the manuscript. All substantive content, empirical analyses, interpretations, and conclusions are my own. I reviewed and take full responsibility for the final manuscript.

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Appendix

Table A1: Population and Amazon's Presence

	All commuting zones (1)	Receive Amazon facility (2)	Never receive facility (3)	Difference (2) – (3) (4)
<i>Panel A. Population</i>				
Total population (thousands)	829 (1,616)	1,219 (1,926)	155 (110)	1,064*** [141]
Aged 18–24 (% of population)	10.9 (3.4)	10.6 (2.7)	11.4 (4.4)	–0.8 [0.5]
White, non-Hispanic (%)	75.2 (16.4)	72.2 (16.2)	80.5 (15.4)	–8.3*** [1.9]
Black (%)	10.2 (11.0)	11.7 (11.1)	7.8 (10.4)	3.9*** [1.3]
Hispanic (%)	9.0 (11.8)	10.6 (13.0)	6.1 (8.5)	4.6*** [1.3]
Foreign-born (%)	6.5 (6.2)	7.9 (6.9)	4.1 (3.7)	3.7*** [0.6]
Urban (% of population)	68.1 (16.8)	75.2 (14.1)	55.8 (13.7)	19.4*** [1.7]
<i>Panel B. Amazon's presence in treated commuting zones</i>				
Year first facility opened		2018.6 (3.8)		
Facilities in 2024 (median: 2)		6.3 (9.3)		
Facility floor space in 2024 (million sq. ft.)		2.04 (3.47)		
Commuting zones	297	188	109	

Notes: Each commuting zone's characteristics are averaged over 2005–2009, before any commuting zone in the sample is treated, and the table reports the mean across commuting zones, with standard deviations in parentheses; commuting zones are not weighted by population. Column (2) contains commuting zones where Amazon opened at least one facility between 2010 and 2024, column (3) the remaining commuting zones in the balanced 2005–2024 panel. Column (4) reports the difference in means, with its standard error in brackets; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Total population and the population aged 18–24 are from published ACS 1-year county estimates; race, ethnicity, and foreign birth are from IPUMS ACS microdata (2005–2009); urban share is from the 2010 Census. Panel B refers to commuting zones that receive a facility; the year of the first opening, the number of facilities, and their floor space in 2024 are from MWPVL International.

Table A2: Balance in Pre-Treatment Trends: Early versus Later Amazon Entry

	Year of first Amazon facility			
	2010–2013 (1)	2014–2016 (2)	2017–2019 (3)	2020–2021 (4)
High school or more (%)	–0.085 (0.067)	–0.060** (0.030)	–0.034 (0.027)	–0.012 (0.032)
Some college or more (%)	0.018 (0.111)	–0.095** (0.037)	–0.030 (0.036)	0.006 (0.044)
Bachelor’s or more (%)	0.025 (0.070)	0.043 (0.033)	0.044 (0.030)	0.051 (0.039)
Master’s or more (%)	0.038 (0.044)	0.032 (0.026)	0.034* (0.020)	0.026 (0.024)
First-time four-year students (% of 18–24)	0.068 (0.076)	0.013 (0.041)	0.011 (0.027)	0.001 (0.029)
College graduates (% of employment)	–0.007 (0.078)	0.129*** (0.039)	0.033 (0.033)	0.011 (0.038)
STEM occupations (% of employment)	0.076* (0.045)	0.011 (0.016)	–0.010 (0.014)	–0.008 (0.013)
Log median household income	0.0016 (0.0022)	–0.0026 (0.0017)	–0.0017 (0.0012)	–0.0021* (0.0012)
Poverty rate (%)	0.081 (0.088)	0.093* (0.050)	0.054 (0.036)	–0.004 (0.042)
Unemployment rate (%)	0.067 (0.068)	0.146*** (0.057)	0.015 (0.025)	0.007 (0.022)
Log population	0.0001 (0.0028)	–0.0012 (0.0024)	–0.0006 (0.0020)	0.0017 (0.0016)
Pre-treatment years	2005–2008	2005–2012	2005–2015	2005–2018
Commuting zones in the group	25	29	39	53
Comparison commuting zones	163	134	95	42

Notes: Each cell reports the difference in the average annual change of the variable during the pre-treatment years, between the commuting zones in the group and the comparison commuting zones, following [Marcus and Sant’Anna \(2021\)](#) and [Cunningham \(2024\)](#). A difference of zero means the two groups were on the same trend before Amazon’s entry. Each column compares commuting zones whose first Amazon facility opened in the indicated years with commuting zones that received their first facility later, which serve as their comparison group in the difference-in-differences design, using only years before the group is treated (treatment begins the year before the first opening). The sample is the balanced 2005–2024 panel (2020 not available) of commuting zones that receive an Amazon facility; the comparison group for the 2020–2021 group consists of commuting zones first entered in 2022–2024. Standard errors, allowing for unequal variances, in parentheses. Attainment and population are from published ACS 1-year estimates for counties with at least 65,000 residents, aggregated to commuting zones; income, poverty, unemployment, and workforce measures are from IPUMS ACS microdata; first-time four-year students (local-draw colleges, as a percentage of the population aged 18–24) are from IPEDS, available from 2006. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: Balance in Pre-Treatment Levels: Early versus Later Amazon Entry

	Year of first Amazon facility			
	2010–2013 (1)	2014–2016 (2)	2017–2019 (3)	2020–2021 (4)
High school or more (%)	–1.49 (0.92)	0.41 (0.71)	0.77 (0.77)	–0.72 (0.95)
Some college or more (%)	0.41 (1.57)	3.42*** (1.28)	2.37** (1.19)	0.67 (1.45)
Bachelor's or more (%)	2.41 (1.48)	5.77*** (1.41)	3.07*** (1.11)	0.68 (1.24)
Master's or more (%)	0.81 (0.69)	2.27*** (0.72)	1.38** (0.59)	0.35 (0.60)
First-time four-year students (% of 18–24)	–1.27*** (0.39)	–1.10** (0.47)	–0.82 (0.56)	0.03 (0.79)
College graduates (% of employment)	3.22** (1.41)	7.65*** (1.32)	4.23*** (0.96)	1.30 (0.98)
STEM occupations (% of employment)	1.00** (0.46)	1.86*** (0.40)	0.90*** (0.29)	0.68*** (0.24)
Log median household income	0.12*** (0.04)	0.18*** (0.03)	0.09*** (0.03)	0.03 (0.03)
Poverty rate (%)	–1.48*** (0.55)	–2.29*** (0.54)	–1.18* (0.65)	0.06 (0.80)
Unemployment rate (%)	0.04 (0.26)	0.41 (0.33)	0.36 (0.29)	0.29 (0.33)
Log population	1.12*** (0.22)	1.71*** (0.17)	1.14*** (0.12)	0.49*** (0.15)
Pre-treatment years	2005–2008	2005–2012	2005–2015	2005–2018
Commuting zones in the group	25	29	39	53
Comparison commuting zones	163	134	95	42

Notes: Each cell reports the difference in the average level of the variable during the pre-treatment years between the commuting zones in the group and the comparison commuting zones. Each column compares commuting zones whose first Amazon facility opened in the indicated years with commuting zones that received their first facility later, which serve as their comparison group in the difference-in-differences design, using only years before the group is treated (treatment begins the year before the first opening). The sample is the balanced 2005–2024 panel (2020 not available) of commuting zones that receive an Amazon facility; the comparison group for the 2020–2021 group consists of commuting zones first entered in 2022–2024. Standard errors, allowing for unequal variances, in parentheses. Attainment and population are from published ACS 1-year estimates for counties with at least 65,000 residents, aggregated to commuting zones; income, poverty, unemployment, and workforce measures are from IPUMS ACS microdata; first-time four-year students (local-draw colleges, as a percentage of the population aged 18–24) are from IPEDS, available from 2006.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A4: Effects of Amazon's Entry on Educational Attainment by Gender and Race

<i>Panel A. By gender</i>				
	Bachelor's+ men (1)	Bachelor's+ women (2)	Master's+ men (3)	Master's+ women (4)
Amazon Entry	0.531** (0.211)	0.541*** (0.192)	0.531*** (0.130)	0.316** (0.140)
Mean Dep. Var.	28.17	27.84	10.61	10.00
Commuting zones	188	188	188	188
<i>Panel B. Bachelor's degree or more, by race and ethnicity</i>				
	White non-Hispanic (1)	Black (2)	Hispanic (3)	
Amazon Entry	0.545** (0.225)	0.670 (0.658)	1.036 (0.707)	
Mean Dep. Var.	31.33	16.70	13.20	
Commuting zones	187	53	47	

Notes: Dependent variables are the percentage-point shares of men or women (Panel A) or of adults in the specified racial or ethnic group (Panel B) aged 25 and older with at least the specified degree, from published American Community Survey 1-year estimates for counties with at least 65,000 residents (2005–2024; 2020 is not available), aggregated to 2020 commuting zones. Estimates are the doubly-robust staggered difference-in-differences estimator of [Callaway and Sant'Anna \(2021\)](#), aggregated over all post-entry years (simple ATT), comparing treated commuting zones to not-yet-treated controls, in the balanced panel of commuting zones that receive an Amazon facility between 2010 and 2024; a commuting zone is treated from the year before its first facility opens. In Panel B, each column excludes commuting zones in which the Census Bureau suppresses any county's estimate for the group in any year; the estimate for Asian adults, available for 11 commuting zones, and the event-study figures are in Online Appendix Section [OA1.1](#). Standard errors from the multiplier bootstrap (1,000 draws), clustered at the commuting-zone level, in parentheses. Mean Dep. Var. is the pre-treatment mean in treated commuting zones. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Effects of Amazon's Entry on First-Time College Enrollment

	Two-Year Colleges (1)	Four-Year Colleges (2)	Public Four-Year (3)	Full-Time First-Time Four-Year (4)
<i>Panel A. Local-draw colleges</i>				
Amazon Entry	−0.057 (0.095)	0.262* (0.134)	0.315** (0.156)	0.199** (0.100)
Mean Dep. Var.	3.88	5.08	4.94	4.78
Commuting zones (treated)	415 (185)	325 (177)	281 (158)	324 (176)
<i>Panel B. All colleges</i>				
Amazon Entry	−0.074 (0.091)	0.173 (0.139)	0.217 (0.151)	0.141 (0.095)
Mean Dep. Var.	4.03	6.96	5.54	6.51
Commuting zones (treated)	422 (187)	377 (184)	314 (169)	377 (184)

Notes: Dependent variable is first-time degree- or certificate-seeking undergraduate enrollment (column 4: full-time first-time enrollment) in the specified type of institution located in the commuting zone, as a percentage of the commuting zone's population aged 18–24, 2010–2023. Enrollment is from IPEDS Fall Enrollment (one all-students record per institution-year) aggregated to 2020 commuting zones through institution counties; population is from IPUMS ACS 1-year microdata. Two-year colleges are institutions that do not award bachelor's degrees and four-year colleges are bachelor's-granting institutions, following [Charles et al. \(2018\)](#). Local-draw colleges exclude institutions whose share of first-time students from other states or abroad was 30 percent or more in 2006 and 2008 (IPEDS residence data), measured before any commuting zone is treated. A commuting zone is treated from the year before its first Amazon facility opens. Estimates are the doubly-robust staggered difference-in-differences ATT of [Callaway and Sant'Anna \(2021\)](#), aggregated over all post-entry years (simple ATT); the estimator differences out time-invariant commuting-zone characteristics and common year shocks by comparing changes in treated commuting zones with contemporaneous changes in comparison commuting zones. Both panels use not-yet-treated commuting zones as comparisons. The sample for each column contains commuting zones with positive enrollment of the relevant type in every year; cohorts treated in 2010 have no pre-period and do not contribute. Standard errors from the multiplier bootstrap (1,000 draws), clustered at the commuting-zone level, in parentheses. Mean Dep. Var. is the pre-treatment mean in treated commuting zones. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: Household Income and the Education Effects

<i>Panel A. Household income, employment, and rents</i>						
	Log median household income (1)	Log mean wage (2)	Poverty rate (3)	Unemployment rate (4)	Labor force participation (5)	Log median gross rent (6)
Amazon Entry	0.015*** (0.006)	−0.000 (0.006)	0.029 (0.164)	−0.311* (0.164)	0.375*** (0.132)	0.039*** (0.007)
Mean Dep. Var.	\$35,470	\$27,058	13.44	7.09	63.16	\$534
Commuting zones	204	204	204	204	204	204
<i>Panel B. By baseline poverty rate</i>						
	Poorer commuting zones	Richer commuting zones	Difference (1) − (2)			
Log median household income	−0.000 (0.008)	0.030*** (0.007)	−0.031*** (0.011)			
Bachelor's or more, adults 25+ (pp)	0.250 (0.356)	0.603** (0.243)	−0.353 (0.431)			
Commuting zones (attainment)	95	93				

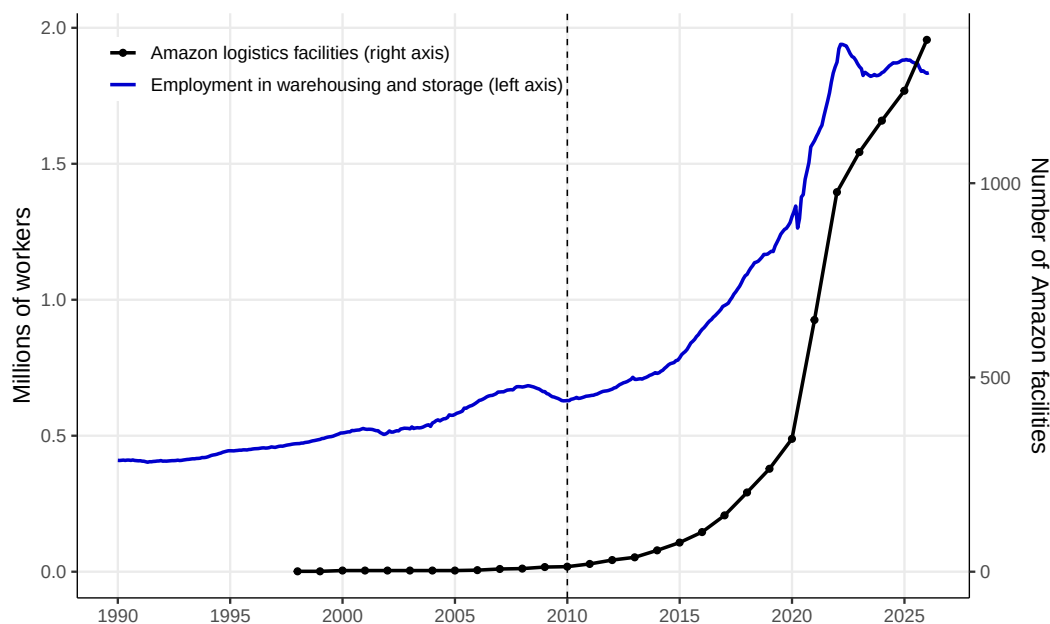
Notes: Each cell is the doubly-robust staggered difference-in-differences estimate of [Callaway and Sant'Anna \(2021\)](#), aggregated over all post-entry years (simple ATT), comparing treated commuting zones with not-yet-treated commuting zones; a commuting zone is treated from the year before its first Amazon facility opens. Panel A: outcomes from IPUMS ACS 1-year microdata aggregated to 2020 commuting zones, 2005–2024; income, wages, and rents in constant 1999 dollars; columns (1), (2), and (6) in logs, columns (3)–(5) in percentage points; Mean Dep. Var. is the pre-treatment mean (geometric mean for logs). Panel B: treated commuting zones are split at the median of the baseline (2005–2009) poverty rate (12.1 percent); poorer commuting zones are above the median. Each cell is estimated within the subsample, with not-yet-treated commuting zones in the same subsample as comparisons. Household income is from IPUMS ACS microdata; the bachelor's share is from published ACS 1-year estimates for counties with at least 65,000 residents. Column (3) tests the difference assuming independent subsamples. Standard errors from the multiplier bootstrap (1,000 draws), clustered at the commuting-zone level, in parentheses. Full estimates are in Appendix Tables [A7](#) and [A8](#). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A7: Effects of Amazon’s Entry on Household Income, Employment, and Rents, 2005–2024

	Log median household income (1)	Log mean wage (2)	Poverty rate (3)	Unemployment rate (4)	Labor force participation (5)	Log median gross rent (6)
Average	0.015*** (0.006)	−0.000 (0.006)	0.029 (0.164)	−0.311* (0.164)	0.375*** (0.132)	0.039*** (0.007)
Mean Dep. Var.	\$35,470	\$27,058	13.44	7.09	63.16	\$534
Commuting zones	204	204	204	204	204	204

Notes: Dependent variables are computed from IPUMS ACS 1-year microdata aggregated to 2020 commuting zones, 2005–2024. Median household income and median gross rent are weighted medians across households (renters for rent) in constant 1999 dollars; the mean wage is the average wage and salary income of workers with positive earnings, in constant 1999 dollars; the poverty rate is the share of persons below the federal poverty line; unemployment and labor force participation rates refer to persons aged 16 and older. Columns (1), (2), and (6) are in logs, so coefficients are approximately proportional changes; columns (3)–(5) are in percentage points. A commuting zone is treated from the year before its first Amazon facility opens. Estimates are the doubly-robust staggered difference-in-differences ATT of [Callaway and Sant’Anna \(2021\)](#) on the balanced panel of commuting zones that receive an Amazon facility, with not-yet-treated commuting zones as comparisons. The estimates are simple ATTs, which average all post-entry group-time effects. Standard errors from the multiplier bootstrap (1,000 draws), clustered at the commuting-zone level, in parentheses. Mean Dep. Var. is the pre-treatment mean in treated commuting zones, in 1999 dollars for columns (1), (2), and (6) (geometric mean). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure A1: The Growth of Warehousing Employment and of Amazon’s Logistics Network



Notes: The blue line (left axis) plots monthly U.S. employment in warehousing and storage (NAICS 493), in millions of workers, from the Bureau of Labor Statistics Current Employment Statistics (series CES4349300001, seasonally adjusted), January 1990 to February 2026. The black line (right axis) plots the cumulative number of Amazon logistics facilities opened in the United States, including fulfillment centers, sortation centers, delivery stations, and air hubs, from MWPVL International records, measured at the end of each year, 1997–2025. The dashed line marks 2010.

Table A8: Effects of Amazon's Entry by Baseline Poverty and Income

	Poorer commuting zones (1)	Richer commuting zones (2)	Difference (1) – (2) (3)
<i>Panel A. Split by baseline poverty rate</i>			
Log median household income	–0.000 (0.008)	0.030*** (0.007)	–0.031*** (0.011)
Bachelor's or more, adults 25+ (pp)	0.250 (0.356)	0.603** (0.243)	–0.353 (0.431)
First-time four-year enrollment	–0.014 (0.193)	0.504** (0.212)	–0.518* (0.287)
Part-time share of enrollment (pp)	–1.302** (0.526)	–1.845 (1.154)	0.543 (1.268)
Commuting zones (attainment sample)	95	93	
<i>Panel B. Split by baseline median household income</i>			
Log median household income	–0.004 (0.007)	0.043*** (0.010)	–0.047*** (0.013)
Bachelor's or more, adults 25+ (pp)	–0.058 (0.290)	0.450 (0.514)	–0.508 (0.590)
First-time four-year enrollment	0.177 (0.166)	–0.043 (0.129)	0.220 (0.211)
Part-time share of enrollment (pp)	–0.610 (0.626)	–3.040** (1.382)	2.430 (1.517)
Commuting zones (attainment sample)	88	100	

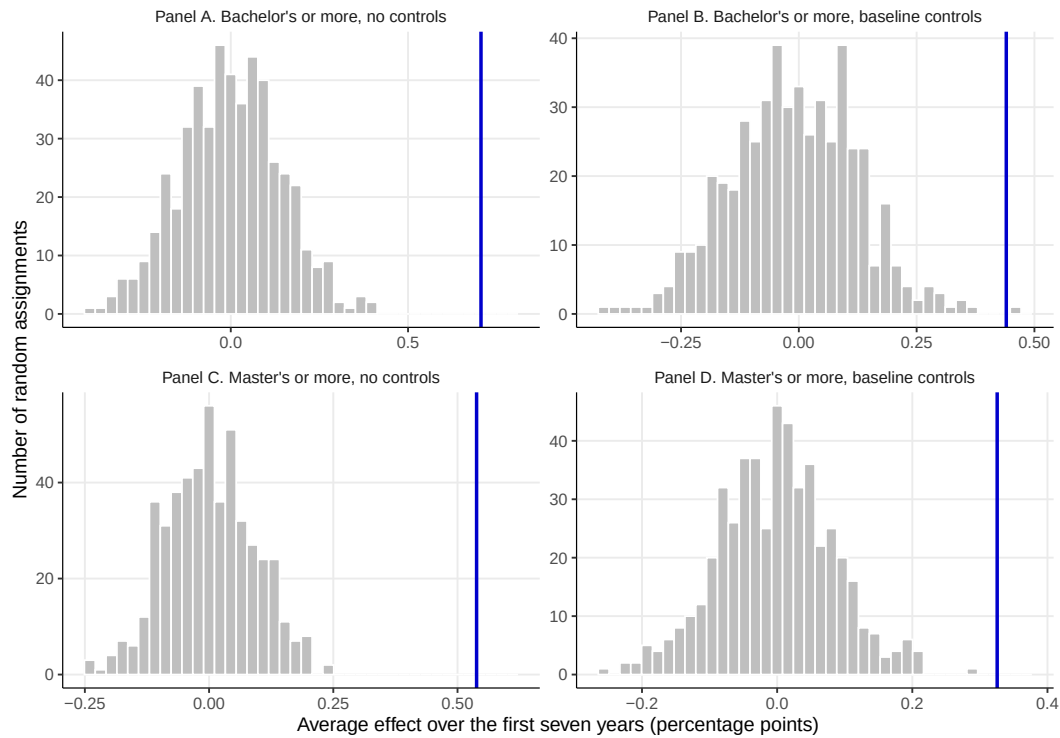
Notes: Commuting zones that receive an Amazon facility are split at the median of the baseline (2005–2009) poverty rate (Panel A; median 12.1 percent) or of baseline median household income (Panel B). Poorer commuting zones are those above the median poverty rate or below the median income; 79 percent of high-poverty commuting zones are also low-income. Each cell is a separate doubly-robust staggered difference-in-differences estimate (Callaway and Sant'Anna, 2021) within the subsample, with not-yet-treated commuting zones in the same subsample as comparisons, aggregated over all post-entry years (simple ATT). Household income is from IPUMS ACS microdata, 2005–2024; attainment is from published ACS 1-year estimates for counties with at least 65,000 residents, 2005–2024 (2020 not available); first-time four-year enrollment at local-draw colleges (as a percentage of the population aged 18–24) and the part-time share of enrollment are from IPEDS, 2010–2023. Column (3) tests the difference assuming independent subsamples. Standard errors from the multiplier bootstrap (1,000 draws), clustered at the commuting-zone level, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A9: Effects of Amazon's Entry on Industry Composition and Establishments

	Amazon Entry (1)	Std. error (2)	Mean before entry (3)
<i>Panel A. Industry share of employment (pp), County Business Patterns, 2005–2023</i>			
Professional, scientific & technical	0.179**	(0.084)	5.03
Information	0.096	(0.069)	2.16
Finance & insurance	0.092**	(0.044)	4.47
Management of companies	0.117**	(0.049)	1.91
Construction	0.157**	(0.066)	5.54
Retail trade	−0.115*	(0.065)	14.74
Accommodation & food	−0.220***	(0.085)	11.17
Administrative & support	−0.139*	(0.083)	5.65
Manufacturing	−0.181*	(0.093)	12.65
<i>Panel B. Establishments, Business Dynamics Statistics, 2005–2023</i>			
Professional, scientific & technical (log)	0.039***	(0.007)	1,054
Information (log)	0.027***	(0.009)	199
Finance & insurance (log)	0.017***	(0.005)	799
Manufacturing (log)	0.004	(0.007)	501

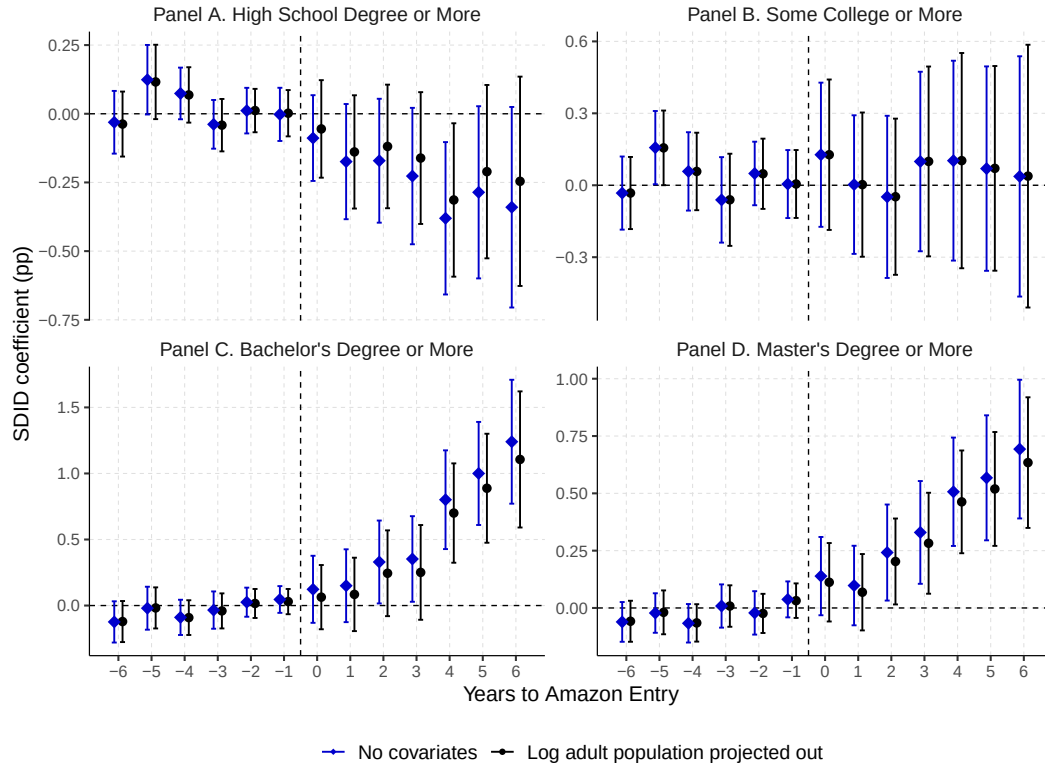
Notes: Panel A: employment in each 2-digit NAICS sector as a percentage of total employment, from County Business Patterns (mid-March employment by workplace) aggregated from counties to commuting zones; employment withheld for disclosure before 2017 is set to the midpoint of the reported size class, and commuting zones whose county composition changes over time are excluded. Panel B: log number of establishments by sector, from the Census Bureau's Business Dynamics Statistics aggregated from counties to commuting zones. A commuting zone is treated from the year before its first Amazon facility opens. Each row is a separate doubly-robust staggered difference-in-differences estimate (Callaway and Sant'Anna, 2021) on the balanced panel of commuting zones that receive an Amazon facility, with not-yet-treated commuting zones as comparisons; column (1) reports the simple ATT, which averages all post-entry group-time effects, and column (2) reports standard errors from the multiplier bootstrap (1,000 draws), clustered at the commuting-zone level. Column (3) is the pre-treatment mean in treated commuting zones (for log outcomes, the geometric mean count). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure A2: Randomization Inference: Actual Estimates versus Random Assignments of Amazon's Entry



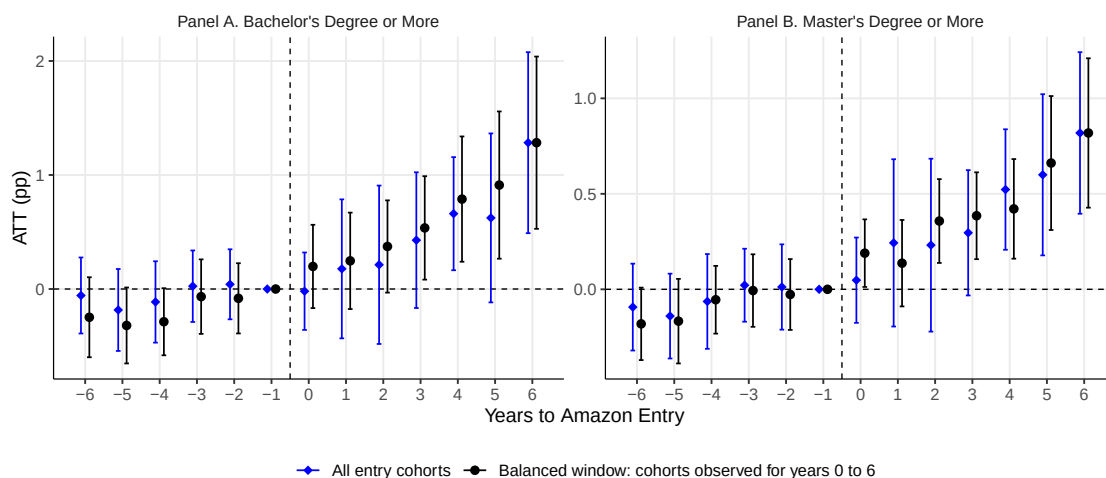
Notes: Each histogram shows the distribution of the simple ATT across 500 random reassignments of treatment years among commuting zones that receive an Amazon facility, holding fixed the number of commuting zones treated in each year. The vertical line marks the actual estimate. Estimates are the doubly-robust staggered difference-in-differences estimates of [Callaway and Sant'Anna \(2021\)](#) with not-yet-treated commuting zones as comparisons; Panels B and D condition on log population in 2005–2009 and the change in log population between 2005–2006 and 2008–2009. Outcomes are the percentages of adults aged 25 and older with at least a bachelor's or a master's degree, from IPUMS ACS microdata, 2005–2024.

Figure A3: Synthetic Difference-in-Differences Estimates: Effects of Amazon’s Entry on Educational Attainment, 2005–2024



Notes: Each panel plots staggered synthetic difference-in-differences event-study estimates (Arkhangelsky et al., 2021) for the percentage of adults aged 25 and older with at least the specified level of schooling, from published American Community Survey 1-year estimates for counties with at least 65,000 residents (2005–2024; 2020 is not available), aggregated to 2020 commuting zones. Estimates at negative event times are placebo effects for the six years before treatment. Blue diamonds: no covariates; black circles: log adult population projected out. Comparison units are commuting zones that never receive an Amazon facility, reweighted to match the pre-treatment path of treated commuting zones. Bars show 95% confidence intervals from 200 bootstrap replications. The sample contains commuting zones observed in every year. Treatment begins the year before Amazon’s first facility opens.

Figure A4: Event-Study Estimates with a Balanced Event Window



Notes: Each panel plots event-time coefficients from the estimator of [Callaway and Sant'Anna \(2021\)](#) with not-yet-treated comparisons, for all entry cohorts (blue diamonds) and for a balanced event window that keeps only cohorts observed for all of event years 0 to 6 (black circles). Panels A and B use the published American Community Survey 1-year county estimates (2005–2024; 2020 is not available); the balanced window keeps the 93 commuting zones whose first facility opened by 2019. Bars show uniform 95% confidence bands from the multiplier bootstrap (1,000 draws), clustered at the commuting-zone level. Event time -1 is normalized to zero.